

REVIEW ARTICLE

Jumping droplets

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Abstract

When microdroplets with quasi-spherical contact angles coalesce together on a low-adhesion substrate, the capillary-inertial expansion of the liquid bridge induces a dramatic out-of-plane jumping event due to symmetry breaking. From the onset of merging, droplet jumping initiates after a capillary-inertial time scale of $t_{ci} \sim 1\text{--}100\ \mu\text{s}$ with characteristic jumping speeds of order $v_j \sim 0.1\text{ m/s}$. This coalescence-induced jumping-droplet effect is most commonly observed among a population of growing dew droplets on a superhydrophobic condenser, but can also occur by colliding deposited droplets together or during droplet sliding on fog harvesters. In this review, we cover the historical development of capillary-inertial jumping droplets, summarize the decade-long effort to rationalize the ultra-low energy conversion efficiency and critical droplet size of the phenomenon, and then present 15 variations on a theme of jumping. Capillary-inertial jumping droplets are not only a visceral illustration of the surprising power of surface tension at the microscale but they also have the potential to enhance phase-change heat transfer, enable self-cleaning surfaces, combat frost formation, harvest energy, and govern the rate of disease spread for wheat crops.

INTRODUCTION

This review paper on jumping droplets parallels the structure of Beethoven's "Variations and Fugue for Piano in E♭ major" (Opus 35). Beethoven deceptively opens the work with several developments of the bass line, rather than the main theme itself. The main theme (later adapted for his landmark third symphony) is then dramatically revealed, followed by 15 clever variations, culminating in a finale comprised of fugal elements that stitch multiple ideas together.¹

Here, our analogous baseline will be developed by tracing the arc of history for capillary-inertial jumping droplets and providing an account of the detective work leading to the breakthrough 2009 report.² The main theme centers on a graph of the jumping velocity as a function of droplet size and the associated mysteries of why the jumping efficiency is so low and why there is a critical minimum size

for jumping. Having established the main theme of the mechanics of coalescence-induced jumping droplets, we will proceed by reviewing 15 variations on a theme of jumping that deal with various subtopics. Variations 1–5 center around subtopics related to phase-change heat transfer, such as measuring the condensation heat transfer coefficient and using electric fields to enhance heat transfer. Variations 6–10 deal with subtopics related to the hydrodynamics of jumping, such as enhanced jumping from macrostructures, directional jumping, and single-droplet jumping that exploits Laplace pressures rather than coalescence. Variations 11–15 deal with nonconventional scenarios and emerging topics, such as jumping from smooth fibers and jumping bubbles. Finally, our conclusion weaves the disparate story lines together, in a scientific rendition of Beethoven's closing fugue. We hope that this review imparts fundamental and practical insights to researchers interested in

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capillary-inertial propulsion at the micro-scale; additionally, we heartily recommend earlier reviews that offer complementary perspectives on the topic.^{3–5}

A BRIEF HISTORY OF JUMPING DROPLETS

Ballistospore ejection

As with most discoveries of unexpected phenomena, it turns out that nature did it first. In at least 30,000 species of mushrooms, yeasts, and pathogenic rusts, spore launching is achieved by exploiting an asymmetric coalescence event to rapidly change its center of mass.^{6–8} Spores that are ejected in this fashion are termed ballistospores. Specifically, coalescence occurs between a dangling droplet preferentially condensed at the hygroscopic base of the ballistospore, called a Buller's droplet,⁹ and a film-like flattened droplet on the adaxial side of the spore itself (Figure 1a).¹³ When the Buller's droplet grows to a characteristic diameter of about $D_c \approx 10 \mu\text{m}$, it is able to merge with the wetted film, which releases a surface tension force of $F_\gamma \approx 1 \mu\text{N}$, sufficient to overcome the rupture force $F_r \approx 0.15 \mu\text{N}$ to fracture the helum that connects the spore to its sterigma.¹⁴ The resulting launch of the ballistospore (first captured with high-speed imaging by Pringle et al.⁶) occurs in a mere $10 \mu\text{s}$, with typical accelerations of $\sim 10,000g$ and initial velocities of $\sim 1 \text{ m/s}$.^{6,15} Not only are ballistospores a natural prelude to jumping-droplet condensation but also their clever use of geometry and local hygroscopic chemicals to produce a combination of dropwise condensation (Buller's droplet) and filmwise liquid (wetted spore) is amazingly reminiscent of recent trends in the literature of using physical and chemical patterns to tailor hybrid and/or patterned modes of condensation.^{16–19}

Mercury droplets jumping from a smooth condenser

While jumping-droplet condensation is now popularly associated with water on superhydrophobic surfaces,² the coalescence-induced launching of mercury from a smooth condenser was discovered in 1969 by Kollera and Grigull, a full four decades prior (Figure 1b).¹⁰ Written in German, an English translation of the experimental setup and discovery is provided here: "In our own experiments, mercury condensed on a vertical tube made of chromium steel with a diameter of 16 mm and a length of 120 mm. During the condensation experiments, it was observed that several of the growing droplets often spontaneously coalesced to form a larger drop on the cooling surface, which then jumped off perpendicularly to the cooling surface. This process was filmed at 44 frames per second and 5X magnification and at 1200 frames per second and 0.5X magnification. The largest of the ejected drops have a diameter of about 0.4 mm."

It may be surprising at first that the jumping-droplet effect was possible on a hydrophilic steel condenser. However, the surface

tension of mercury is $\gamma_{\text{Hg}} \approx 486 \text{ mN/m}$, an order of magnitude larger than that of water ($\gamma \approx 73 \text{ mN/m}$). This explains why dropwise condensation with large contact angles ($\theta \approx 150^\circ$) was possible on steel, despite it promoting filmwise condensation for water. Considering when it was published, the quality of the high-speed imaging in this report is truly remarkable and ahead of its time. The theoretical model, while convoluted and limited by modern standards, accurately posited that surface energy was released upon coalescence for partial conversion into kinetic energy. They also recognized that the amount of energy released upon coalescence was a function of the number of droplets and their contact angle. It is regrettable that the lack of an English translation has hampered the awareness of this radically innovative work. Indeed, it was not even cited in the 2009 report of jumping water droplets,² although it was eventually rediscovered and properly cited in key follow-up work.^{20–22}

Discovery of jumping-droplet condensation on superhydrophobic surfaces

Most early reports of condensation on superhydrophobic surfaces used micro-roughness and observed an impaled Wenzel state, such that droplet adhesion was much larger than that of dropwise condensation on a smooth hydrophobic surface.^{23–26} A breakthrough paper, published in 2003 by Lau et al.,¹¹ attempted to grow condensation on a superhydrophobic surface comprised of a vertically aligned carbon nanotube forest (rather than a micro-structure). Environmental scanning electron microscopy (ESEM) images revealed quasi-spherical dew droplets whose contact lines were obviously trapping air pockets beneath in the low-adhesion Cassie state (Figure 1c). The ESEM imaging also revealed that the carbon nanotubes beneath droplets experienced elastocapillary bundling, a topic that has received considerable attention in the two preceding decades.²⁷ However, the resulting dynamic behavior of these spherical dew droplets was not reported, which brings us to the next landmark work.

Chen et al.,¹² published a paper in 2007 that foreshadowed the jumping-droplet effect. This report compared dropwise condensation on one-tier versus two-tier superhydrophobic surfaces. The microscale roughness was comprised of an array of silicon micropillars, while the nanoscale roughness was a forest of carbon nanotubes. Condensation was observed on three different types of surface structures: micropillars only, nanotubes only, or two-tier with the nanotubes on top of each micropillar and in the intermediate alleys. When applying a conformal hexadecanethiol coating (intrinsic contact angle of $\theta = 101^\circ$ on an equivalently smooth surface), the impaled Wenzel state was thermodynamically stable for the micropillar-only surface, whereas the air-trapping Cassie state was preferable for the nanotube-only and two-tier surfaces. As a consequence, the dropwise condensate featured noncircular, impaled contact lines for the micropillar-only case, compared to quasi-spherical apparent contact angles and low

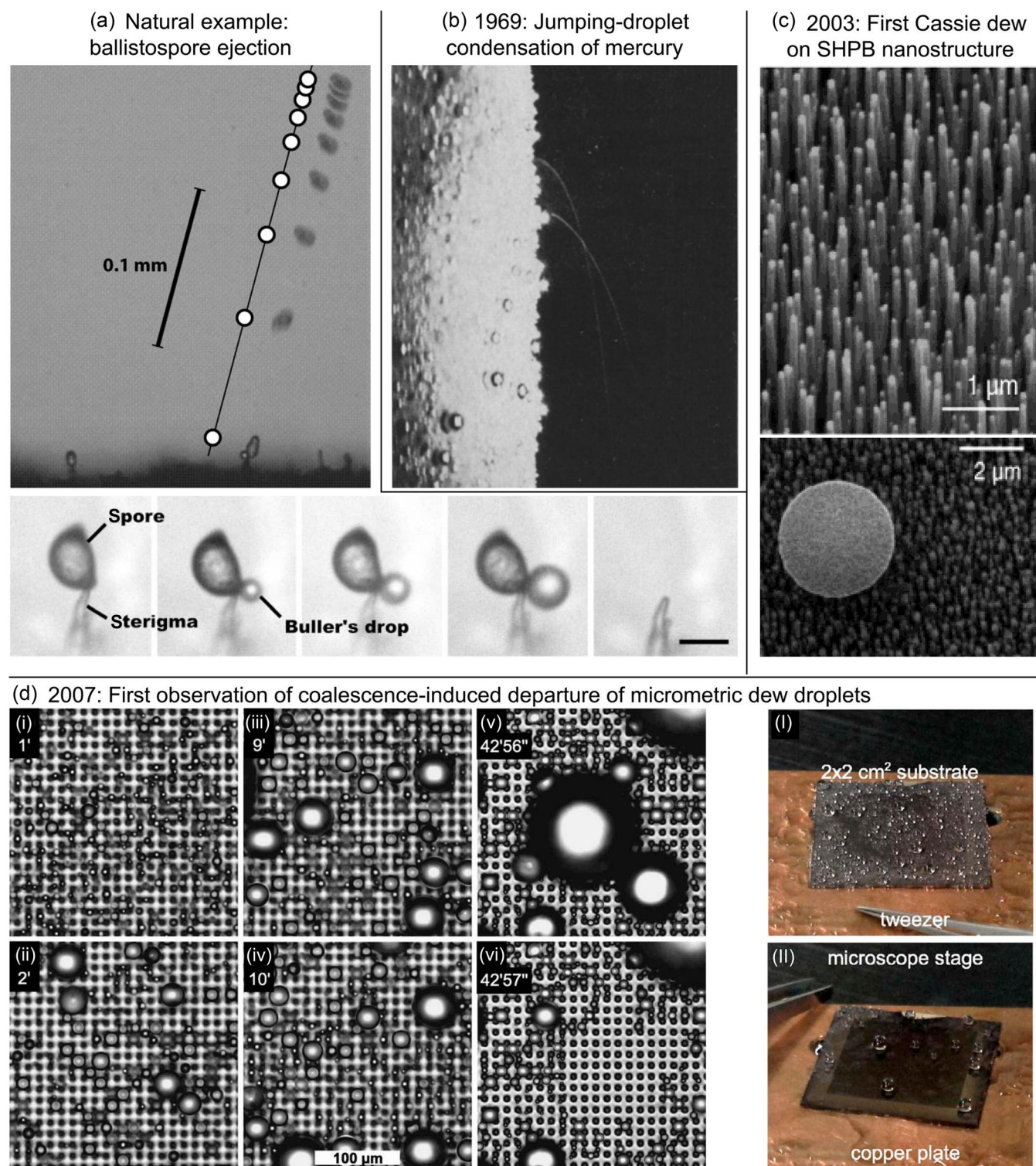


FIGURE 1 Preludes to jumping-droplet condensation. (a) Capillary-inertial launching naturally occurs during ballistospore ejection (top image). A Buller's drop is grown at the hygroscopic tail of a spore, which eventually merges with the wetted top of the spore to detach it from its sterigma (bottom image). Reproduced with permission.⁶ Copyright 2005, the Mycological Society of America. (b) The first report of liquid-only capillary-inertial jumping is from 1969, observed during the condensation of mercury on steel. Reproduced with permission.¹⁰ Copyright 1969, Springer Nature. (c) In 2003, it was discovered that water can condense in a low-adhesion Cassie state when the superhydrophobic surface is nanostructured, rather than microstructured. Reproduced with permission.¹¹ Copyright 2003, the American Chemical Society. (d) Using a two-tier superhydrophobic surface, coalescence-induced departure of dew was observed for the first time in 2007, but the capillary-inertial jumping was not discovered due to the lack of high-speed or side-view imaging. Reproduced with permission.¹² Copyright 2007, AIP Publishing.

droplet adhesion for the surfaces with nanotubes. Up to this point in the paper, the results on the nanostructured surfaces are comparable to those reported by Lau et al.¹¹ The major advancement came from subsequently observing a sudden disappearance of microscale dew droplets during their growth, despite a horizontal surface orientation (Figure 1d, left panel). In particular, it was remarked that “rapid droplet motion, usually triggered by coalescence, is apparent” such that “the condensate drops frequently traveled large distances (0.1–10 mm) over a short period of time (~10 ms) at seemingly random directions.” They also observed that vigorously shaking the substrate, to boost the number of coalescence events and their resulting departure, left the surface “almost completely dry” (Figure 1d, right panel). High-speed imaging was not used, such that the specifics of the

coalescence-induced departure mechanism and subsequent droplet trajectories remained veiled.

Two years later, Boreyko and Chen used the same two-tier surfaces and top-down microscopy, but this time with a high-speed camera (Figure 2a).² Two key observations were made. First, it was confirmed that coalescence was always occurring just before droplet motion, validating that a release of surface energy was the mechanism from an energetic standpoint. Second, it was observed that moving droplets became blurry and tended to leave neighboring droplets undisturbed as they traveled across the surface. Videos were typically captured at 20,000 fps, so this fuzziness was not due to motion blur. Rather, it was deduced that the increasing blurriness during motion was due to an out-of-plane trajectory away from the microscope's focal plane, which simultaneously explains why

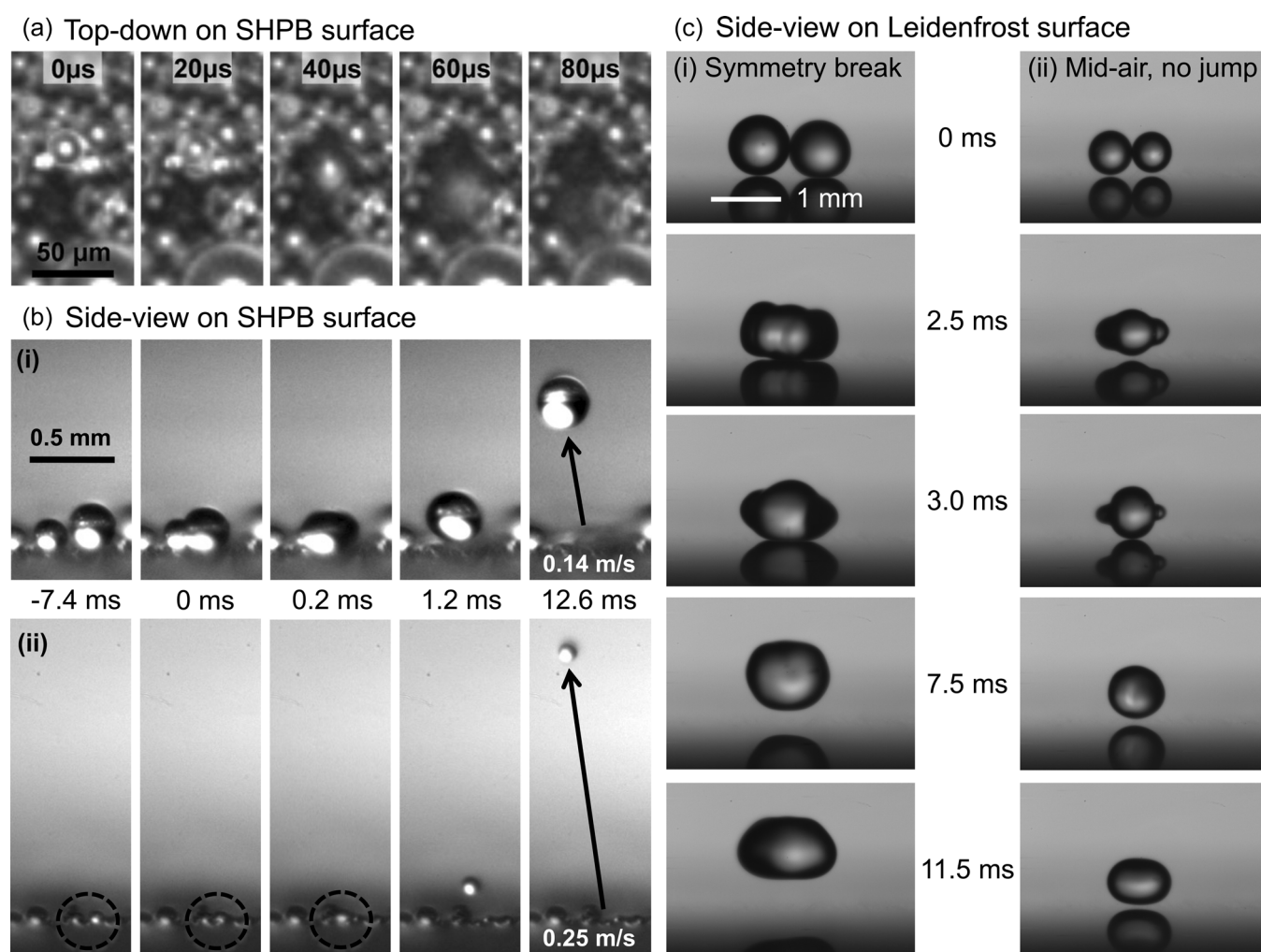


FIGURE 2 Discovery of jumping water droplets. (a) Top-down high-speed imaging revealed that coalescence-induced self-propelled droplets became blurry, suggesting an out-of-plane trajectory. Reproduced with permission.² Copyright 2009, the American Physical Society. (b) First explicit side-view high-speed capture of jumping-droplet condensation; (i) larger micro-droplets jumped slower than (ii) smaller micro-droplets. Reproduced with permission.²⁸ Copyright 2010, AIP Publishing. (c) First imaging of deposited water droplets jumping on a superheated Leidenfrost stage. (i) When coalescence occurred near the substrate, the levitating Leidenfrost droplets jumped analogous to dew on a superhydrophobic surface. This confirms a universal symmetry breaking mechanism for jumping that is applicable to any low-adhesion substrate. (ii) When droplets coalesced in mid-air just above the surface, no symmetry breaking or jumping occurred. Reproduced with permission.²⁸ Copyright 2010, AIP Publishing.

neighboring droplets were typically unperturbed by the lateral component of motion.

Switching the high-speed camera to a side-view microscope, Boreyko was able to explicitly capture the coalescence-induced jumping dynamics for the first time (Figure 2b).² By selecting jumping events that remained in the focal plane throughout, the jumping velocity was able to be measured for the first time. The characteristic jumping size of $D \sim 10\text{--}100\ \mu\text{m}$, initial jumping speed of $v_j \sim 0.1\ \text{m/s}$, and maximum jumping height of $h \sim 1\ \text{mm}$ were all remarkably similar to that of a ballistospore launched by a merging Buller's droplet. It was observed that smaller micro-droplets jumped faster and higher than larger micro-droplets, although this broke down beneath a critical droplet size as will be discussed later. The Reynolds number for a jumping droplet is given by $Re = \rho v_j R_f / \mu$, where ρ and μ are the liquid's density and viscosity, respectively, and R_f is the final radius of the merged droplet. Using high-speed imaging to measure v_j , $Re \sim 1$ (for $R_f \sim 10\ \mu\text{m}$) or $Re \sim 10$ (for $R_f \sim 100\ \mu\text{m}$), confirming that the inertia of jumping is appreciable despite the droplet's micro-scale size. Jumping occurred on both the nanoscale-only border of the surface and the two-tier central portion, validating that nanoscale roughness was the key physical feature and the addition of micro-roughness was optional.²

Jumping droplets on a Leidenfrost surface

Shortly after the discovery of self-propelled droplets on the superhydrophobic surface, Chen hypothesized that the jumping hydrodynamics should be universal for large-angle droplets merging on any low-adhesion substrate. Superior even to superhydrophobic surfaces, the lowest-adhesion surface is a superheated plate, where a droplet levitates on its own water vapor and avoids physical contact with the substrate entirely.²⁹ Sure enough, upon spraying a mist of small droplets onto an aluminum plate heated above the Leidenfrost point of 150°C ,³⁰ the levitating droplets were observed to jump upon coalescing with each other. To better control the effect, Boreyko designed a Leidenfrost droplet collider, where two droplets were trapped behind gates located on opposing ramps.²⁸ By using solenoids to magnetically open both gates simultaneously, the two Leidenfrost droplets rolled down the ramps and the resulting collision and jumping were captured by side-view high-speed imaging (Figure 2c,i). Fortuitously, the droplets tended to show a bouncing stride as they collided into each other. When the droplet collision occurred mid-bounce, no jumping occurred (Figure 2c,ii). In contrast, when the droplets merged in proximity to the substrate, jumping commenced analogous to that observed on the superhydrophobic condenser. These figures and corresponding video footage can be seen in the 2010 report associated with an entry in the 2010 APS Division of Fluid Dynamics Gallery of Fluid Motion.²⁸ This result validated that the substrate only plays a role for jumping in terms of symmetry breaking, where the liquid bridge is free to expand upward, but its downward growth is constrained by the underlying substrate. Thus, the two key criteria required for jumping are large apparent

contact angles, to ensure that the liquid bridge can expand omnidirectionally, and low surface adhesion, to ensure that the merging droplet does not remain stuck to the substrate during symmetry breaking and liftoff.

Back to nature: Self-cleaning

With this newfound awareness of capillary-inertial jumping droplets on large-angle and low-adhesion surfaces, researchers discovered a plethora of natural examples that extended far beyond the original archetype of ballistospores. Wisdom et al. discovered that jumping condensate on superhydrophobic cicada wings (Figure 3a,b) serve to remove encapsulated particles (e.g., pollen grains).³¹ Indeed, it was even shown that a clump of multiple $50\text{-}\mu\text{m}$ diameter glass particles could be dramatically flung from the substrate as vapor preferentially condensed on them (Figure 3c). Similar jumping and self-cleaning effects were subsequently visualized on lotus leaves,³⁴ geckos,³⁵ and water strider legs.³⁶ Interestingly, dew droplets can even remove a particle from a substrate when the droplets are fully confined atop the particle itself.³⁷ A recent work by Yan et al.³⁸ systematically characterized the capillary-inertial jumping of various droplet-particle interactions on butterfly wings, cicada wings, and clover leaves. They found that density, size, and wettability all played a role in the launching dynamics, with rotation occurring in addition to translation during jumping.

In 2019, Nath et al. observed jumping on a common wheat leaf, despite the relatively modest apparent advancing contact of water ($\theta_A^* \approx 127^\circ$).³² On leaves diseased with leaf rust, the jumping droplets were confirmed to launch fungal spores beyond the quiescent boundary layer for wind-assisted dispersal to downstream targets, with as many as ~ 100 spores being dispersed from a single diseased leaf (Figure 3d,e).³³ Collectively, these studies of biological organisms reveal that capillary-inertial droplet jumping can include solid particles and is a double-edged sword in nature. Mushrooms use the phenomenon to sow the next generation, cicadas and geckos expel contamination, but the same effect is potentially responsible for the propagation of plant diseases in staple food crops such as wheat.

TWO CENTRAL MYSTERIES OF JUMPING-DROPLET CONDENSATION: ENERGY INEFFICIENCY AND CRITICAL SIZE

The 2009 report may have unveiled the jumping-droplet effect for water droplets,² but it also raised two central mysteries that took a decade to fully unpack. Both of these puzzles were visually summarized in figure 5 of that letter, replotted here in Figure 4 with additional data from later reports.^{39–43} Normalizing the initial take-off velocity by the idealized capillary-inertial velocity, the ratio is $v^* \approx 0.17\text{--}0.26$, with the minor variance in v^* depending on the particular superhydrophobic surface used for a given report. This

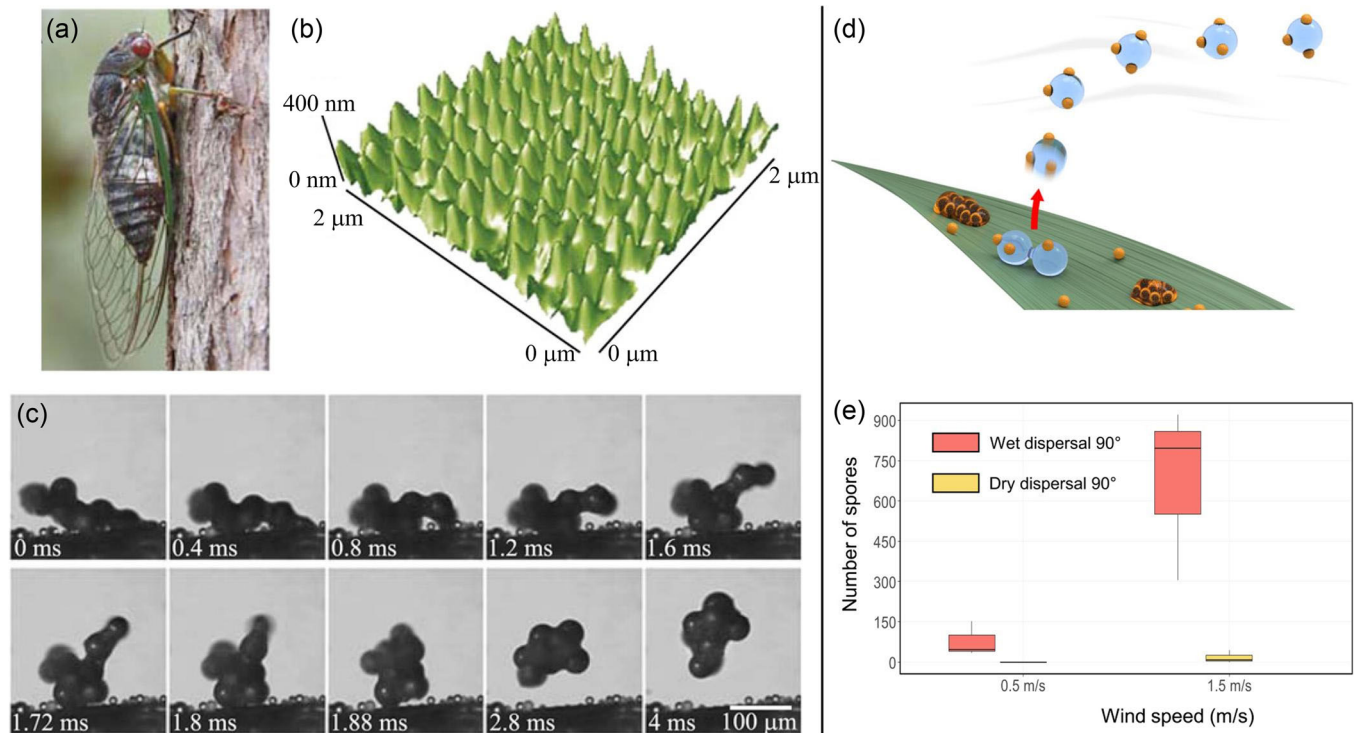


FIGURE 3 Follow-up studies revealed more natural implementations of jumping droplets. (a, b) Photograph and atomic force microscopy of cicada wings. (c) Jumping dew is able to encapsulate and release microparticles from the surface, revealing a self-cleaning mechanism. Reproduced with permission.³¹ Copyright 2013, the National Academy of Sciences. (d) On a wheat leaf infected with rust spores, jumping dew can remove spores but has the unintended consequence of wind-assisted dispersal to downstream leaves. Reproduced with permission.³² Copyright 2021, the National Academy of Sciences. (e) A single infected leaf can disperse hundreds of spores downwind during a dew cycle (pink bars), whereas spores remain largely adhered to dry leaves (yellow bars). Reproduced with permission.³³ Copyright 2021, the National Academy of Sciences.

implies that only about 4–6% of the surface energy released by coalescence is successfully converted into translational kinetic energy (given that it is proportional to the square of the velocity). Hence the first mystery: where does the rest of the released energy go? The second mystery: beneath a critical droplet size, the value of v^* rapidly decreased down to zero, such that jumping did not occur upon coalescence for sufficiently small droplets. This section will review the papers that resolved these questions using ingenious combinations of numerical models, analytical arguments, and nanofabrication.

Inefficiency of jumping

First, let us consider the energetics of converting a coalescence-induced reduction in surface energy into translational kinetic energy. We will use the simplest possible case of two same-sized droplets each showing an initial radius R and an apparent (advancing) contact angle approaching $\theta_A^* \rightarrow 180^\circ$; more complex solutions accounting for size mismatch or finite contact angles are available in the literature.^{2,44–46} The initial surface energy of both droplets combined is given by $E_{s,i} \approx 2(4\pi R^2)\gamma$, where γ is the surface tension of the water–air interfaces. Post coalescence, neglecting hydrodynamic deformation, the surface energy is reduced to $E_{s,f} \approx 4\pi R_f^2\gamma$, where R_f is the final radius of the merged

droplet. From conservation of mass of an incompressible liquid, $2R^3 = R_f^3$, enabling the substitution of $R_f^2 = 2^{2/3}R^2$. The (negative) change in surface energy due to coalescence is given by $\Delta E_s = E_{s,f} - E_{s,i} \approx 4(2^{2/3} - 2)\pi R^2\gamma$. Presuming a perfect conversion into translational kinetic energy, $|\Delta E_s| = E_k \approx \frac{1}{2}\rho \frac{4}{3}\pi (2R^3)v_{ci}^2$, where ρ is the density of water and v_{ci} is the idealized capillary-inertial velocity. Solving for the capillary-inertial jumping speed:

$$v_{ci} \approx \left(\frac{3(2 - 2^{2/3})\gamma}{\rho R} \right)^{1/2} \approx 1.11 \left(\frac{\gamma}{\rho R} \right)^{1/2} \sim \left(\frac{\gamma}{\rho R} \right)^{1/2}. \quad (1)$$

Notice how, for same-sized spherical droplets, the numerical prefactor of 1.11 is very close to unity. It therefore turns out that a crude scaling analysis: $R^2\gamma \sim \rho R^3 v_{ci}^2$, where $R \sim R_f$, would have yielded approximately the same result.

The efficiency of jumping can be gauged by measuring the initial take-off speed (v_j) with a high-speed camera and nondimensionalizing by this idealized capillary-inertial velocity: $v^* = v_j/v_{ci}$. Initiating from the 2009 report but consistent with multiple follow-up reports, $v^* \approx 0.17$ for droplets readily exceeding the critical size where coalescence-induced jumping is possible. In terms of energy conversion efficiency, $\frac{1}{2}mv_j^2 \leq 4\%|\Delta E_s|$ (i.e., $\eta \approx 4\%$), where m is the mass of the jumping droplet, which is astonishingly inefficient considering the minimal surface adhesion of superhydrophobic surfaces.

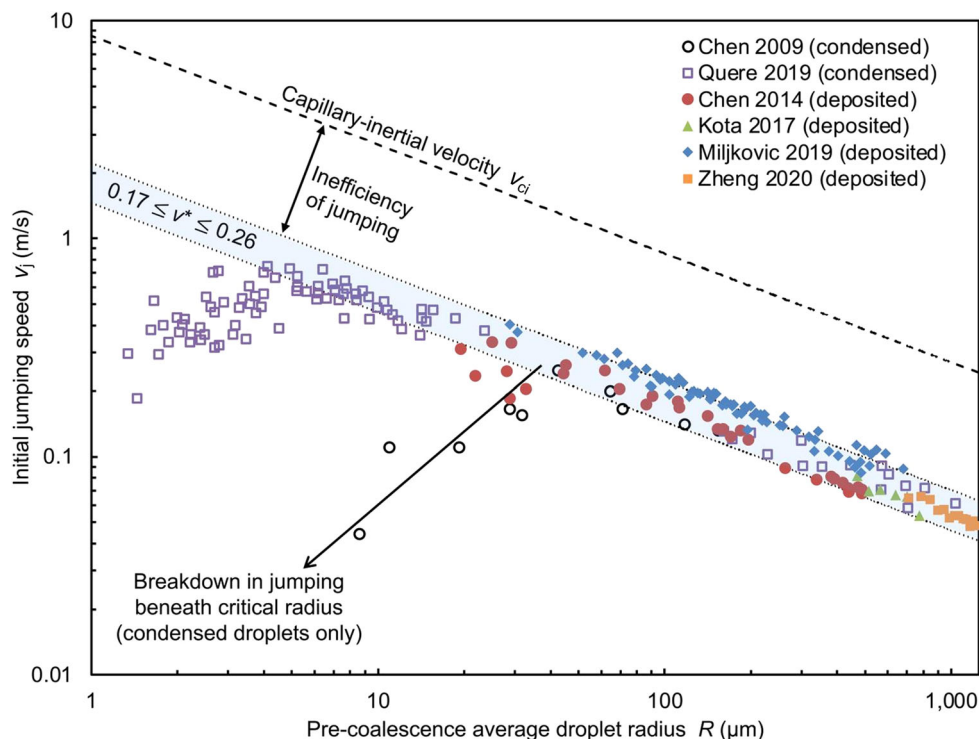


FIGURE 4 Measured values of the take-off speed of jumping droplets as a function of the pre-coalescence droplet radii, for same-sized droplet pairs. Data from six different reports are overlaid onto the graph, with the names in the legend referring to the corresponding author,^{2,39–43} showing a remarkable convergence of results across a variety of superhydrophobic and Leidenfrost surfaces. For deposited droplets (filled symbols), the jumping speeds were parallel to the capillary-inertial velocity curve (dashed black line) but downshifted to a much lower magnitude (blue band). This illustrates the inherent hydrodynamic inefficiency of the capillary-inertial jumping process, where $v^* = v_j/v_{ci} \approx 0.17\text{--}0.26$. Condensed droplets (open symbols) also followed the blue band for larger values of R , but at smaller values, suddenly diverged and plummeted toward $v_j \rightarrow 0$. The main theme of this review paper is to summarize why capillary-inertial jumping is inefficient and, for condensed droplets, features a critical minimum size.

An important side note: the 2009 report normalized experimental measurements of v_j by the version of v_{ci} that contains the numerical prefactor. When accounting for size mismatch but retaining the approximation of $\theta_A \rightarrow 180^\circ$, this results in²:

$$v_{ci} \approx \sqrt{\frac{6\gamma}{\rho} \frac{R_1^2 + R_2^2 - (R_1^3 + R_2^3)^{2/3}}{R_1^3 + R_2^3}}, \quad (2)$$

where R_1 and R_2 are the radii of each droplet before coalescence and $v_{ci} \approx 1.11(\gamma/(\rho R))^{1/2}$ is recovered when $R_1 = R_2$. The advantage of this approach is that the energy conversion efficiency is simply expressed as $\eta = v^*$, and the disadvantage is that the characteristic velocity scale (i.e., v_{ci}) varies from trial to trial. Conversely, virtually all follow-up reports nondimensionalized v_j by the generic scaling version of v_{ci} , which drops the numerical prefactor. This benefits from a universal capillary-inertial velocity scale but does require a numerical prefactor if an accurate estimation of η is desired (i.e., now $\eta \propto v^*$ rather than $\eta = v^*$). Neither approach is wrong, but care must be taken when extracting η from v_{ci} or when comparing values across multiple reports. This review (including the y-axis of Figure 4) converts to this latter approach, such that the 2009 report would have yielded $v^* \approx 0.19$ rather than $v^* \approx 0.17$.

Why are v^* and η so low? It is tempting to blame viscous dissipation, but let us consider this common assertion critically. The Ohnesorge number is defined as the ratio of the visco-capillary time scale (t_{vc}) to the capillary-inertial time scale (t_{ci}):

$$Oh = \frac{t_{vc}}{t_{ci}} = \frac{\mu R/\gamma}{\sqrt{\rho R^3/\gamma}} = \frac{\mu}{\sqrt{\rho \gamma R}}. \quad (3)$$

For viscosity to be appreciable relative to capillary-inertial effects, an Ohnesorge number of $Oh \geq 0.1$ is required. For viscous dissipation to dominate and subdue capillary-inertial jumping entirely, a critical value of $Oh_c \approx 1$ is satisfied for radii of $R_c \sim 10$ nm or smaller, several orders of magnitude less than the characteristic jumping sizes of $R \sim 10$ μm or $R \sim 100$ μm observed experimentally. Some reports use the expression $v^* = \sqrt{1 - Oh}$, which presumes that viscous dissipation accounts for the inefficiency of the jumping process.^{40,47} However, considering a typical case of $R \approx 100$ μm, this results in $Oh \approx 0.01$ and a prediction of $v^* \approx 0.995$, obviously a far cry from the reality of $v^* \approx 0.2$. Therefore, viscous dissipation cannot be blamed for the poor conversion of released surface energy into translational kinetic energy for microscale droplets.

Another suspect is the work of adhesion inherent to detaching the liquid–solid interfaces from the tops of the surface

roughness during jumping. The work of adhesion scales as $W_{ad} \sim \phi\gamma(1 + \cos\theta)(R \sin\theta_A^*)^2$, where ϕ is the solid fraction of a Cassie droplet's base that directly contacts the surface structure and the droplet's apparent contact radius with the substrate is given by $a = R \sin\theta_A^*$, such that its apparent contact area scales as $A \sim a^2$. The solid fraction of a Cassie droplet suspended on a superhydrophobic surface can range from $\phi \approx 0.02$ to $\phi \approx 0.25$, depending on the geometry of the surface roughness and whether one-tier or two-tier structures are used.¹² Using characteristic values of $\phi \approx 0.1$, $\theta \approx 100^\circ$ (for the intrinsic angle of the hydrophobic coating), $R \approx 100 \mu\text{m}$, and $\theta_A^* \approx 160^\circ$, we obtain $W_{ad} \approx 7 \text{ pJ}$. This pales in comparison to the characteristic release in surface energy upon coalescence: $\Delta E_s \sim \gamma R^2 \sim 700 \text{ pJ}$, which is approximately two orders of magnitude larger. Both of these energy terms are proportional to R^2 , so this large ratio is not subject to change with varying droplet sizes. On two separate occasions, the death of writer Mark Twain was prematurely assumed by newspaper reporters: once when his cousin passed and again when the waters near his (docked) boat became rough. Just as reports of Mark Twain's death have twice been exaggerated, so have reports of the death of droplet jumping due to viscous dissipation or surface adhesion.

Scaling analysis is a useful tool that correctly identified the physical relationship of $v_j = \beta(\gamma/(\rho R))^{1/2}$, where β is a numerical prefactor, but it cannot solve for β . If β were close to unity, this would not be especially bothersome, but $\beta \approx 0.2$ is nontrivial and demands explanation. Numerical simulations were therefore used to fill in

this important knowledge gap. Published between 2013 and 2014, three independent reports by Liu et al.,²¹ Enright et al.,²² and Nam et al.⁴⁸ utilized three-dimensional finite-element simulations to explicitly solve for v^* and η . These simulations used a flat substrate with two same-sized droplets showing a contact angle of 180° , with both gravity and surface adhesion being ignored. The central finding is that coalescence-induced symmetry breaking and subsequent jumping inherently generate large pressure fluctuations that cause capillary waves and dramatic oscillations between oblate and prolate droplet shapes. Most of the released surface energy is therefore being exchanged into oscillatory kinetic energy, with a much smaller fraction being converted into translational kinetic energy. An important but confusing nuance is that the oscillatory kinetic energy itself will slowly dampen over time due to viscous dissipation, but this does not change the fact that during the original coalescence-induced take-off, viscous losses are quite secondary with regard to the low value of v^* .

Figure 5 shows a prototypical simulation from Liu et al.,²¹ where $Oh = 0.00375$, $\mu_l/\mu_g = 12.9$, and $\rho_l/\rho_g = 50$ are the viscosity and density ratios of the liquid droplets to ambient gas, and the corresponding initial droplet radius is $R = 100 \mu\text{m}$ for a hypothetical system temperature of 100°C . (For water droplets and ambient air near room temperature, $Oh = 0.0037$ results in $R \approx 1 \text{ mm}$, which is approaching the upper limit for R due to gravity and therefore represents the lower limit for Oh .) Same-sized coalescence initiated at $t^* = t/t_{ci} = 0$, where $t_{ci} = R/v_{ci}$ is the capillary-inertial time scale used to normalize time. The expanding liquid bridge impacted the

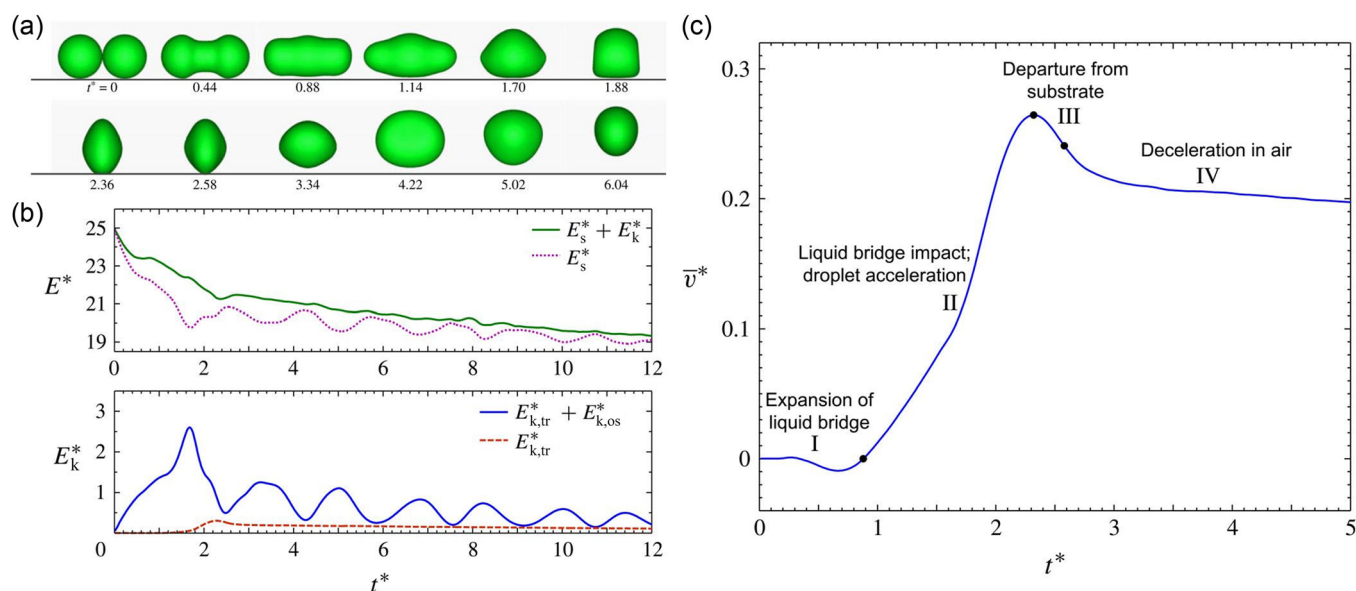


FIGURE 5 Numerical simulations rationalizing the inefficiency of jumping droplets. Reproduced with permission.²¹ Copyright 2014, Cambridge University Press. (a) Side-view phase-field simulations for droplet jumping from a zero-adhesion surface, with $Oh = 0.00375$. The nondimensional time (t^*) is normalized by the capillary-inertial time-scale, with time zero being the onset of coalescence of two same-sized droplets. The results confirm symmetry breaking as the mechanism for jumping and also highlight the dramatic droplet oscillations that occur. (b) Evolution of the nondimensional interfacial surface energy (E_s^*) and the total kinetic energy (E_k^*) during the jumping process shown in (a). The second graph breaks down the kinetic energy in terms of the translation kinetic energy ($E_{k,tr}^*$) and oscillatory kinetic energy ($E_{k,os}^*$), with the latter dominating to explain the inefficiency of jumping. All nondimensional energies are normalized by γR^2 . (c) Variance of the nondimensional jumping velocity over time, revealing four distinct stages of jumping as labeled in the graph.

substrate at $t^* = 0.88$, and the resulting upward acceleration resulted in complete liftoff from the surface by $t^* = 2.58$. During mid-air jumping, the averaged oscillatory kinetic energy is several times larger than the translational kinetic energy. The magnitude of the oscillatory kinetic energy over time is sinusoidal in nature, as a large portion is exchanged back and forth as excess surface energy. Over time, the period-averaged oscillatory kinetic energy is decreasing due to internal viscous dissipation, but appreciable oscillations remain even at $t^* > 10$ due to the low Oh number. Similarly, the translational kinetic energy is weakly decreasing over time even when removing gravity from the simulations, but this is predominantly due to air drag. Repeating this simulation for ever-increasing values of Oh , the value of v^* decreases as expected, while the t^* where liftoff occurs is increasing. By a critical Ohnesorge number of $Oh_c \approx 0.3$ (equivalent to $R \approx 150$ nm for water), the simulations indicated that jumping ceases to occur.

For the capillary-inertial-dominant regime of $Oh < 0.01$, an inherent initial jumping velocity of $v^* \approx 0.23$ is captured by the numerical simulations,^{21,22} in good agreement with the experimental findings summarized in Figure 4. This illustrates the modest importance of contact angle hysteresis and surface adhesion, as the model completely ignores them and yet captures the experimental efficiency. Follow-on numerical simulations accounted for the effects of varying air density,⁴⁹ three-droplet coalescence events,⁵⁰ droplet size mismatch,⁵¹ multihop chain reactions,⁵² and jumping from a biphilic substrate featuring wettability patterns.⁵³

Critical size for jumping

On a superhydrophobic surface, capillary-inertial jumping fails to occur upon coalescence for dew droplets beneath a critical size, $R < R_c$.² At first blush, this is baffling, as the capillary-inertial velocity increases with decreasing droplet size ($v_{ci} \propto R^{-1/2}$). Many reports have blamed this cutoff size on viscous dissipation, but a typical value of $R_c \approx 10$ μm yields a very small Ohnesorge number of $Oh \approx 0.04$,² well below the jumping cutoff of $Oh_c \approx 0.3$.²¹ It follows that viscous dissipation is a false scapegoat for the cutoff size, just as it was for the poor efficiency of v^* for jumping events at $R > R_c$. Using ESEM, Dietz et al.⁵⁴ and Rykaczewski⁵⁵ revealed that the apparent contact angle of a dew droplet increases dramatically during its early growth within the surface structure (Figure 6a). Further, much smaller cutoff sizes of $R_c \approx 1$ μm were observed when using especially tall and slender nanostructures for the surface roughness.^{56–58} Finally, when using deposited droplets on a superhydrophobic surface⁴² or a Leidenfrost stage³⁹ (where inflation is not applicable), the cutoff size for jumping does not exist (see Figure 4). Combining these insights, it is clear that the cutoff size is related to the dew droplets having to inflate to a sufficiently large contact angle and low-adhesion Cassie state for coalescence-induced jumping to be possible.

Figure 6b–d summarizes the inflation model that was pioneered by Wang et al.⁵⁹ and matured by Mouterde et al.⁵⁸ and especially Mulroe et al.⁵⁶ A just-formed liquid embryo is able to successfully

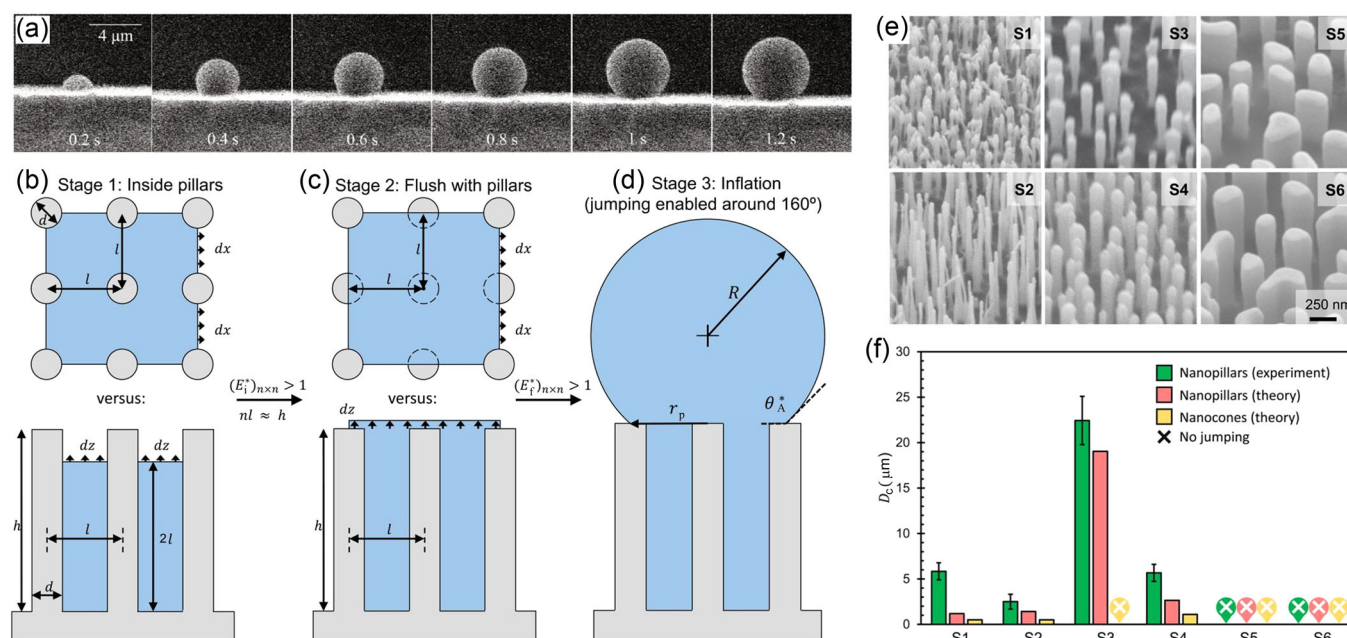


FIGURE 6 Droplet inflation during condensation on superhydrophobic nanostructures. (a) Environmental scanning electron micrographs of a dew droplet inflating into a mature Cassie droplet during early-time growth. Reproduced with permission.⁵⁵ Copyright 2012, the American Chemical Society. (b–d) Three-stage inflation model for a liquid embryo nucleating within an ordered nanostructure. (e, f) Experimental confirmation of the inflation model. The critical minimum diameter for same-sized droplets to jump upon coalescing was compared to that predicted by inflation theory. Aside from surface S1, where elastocapillary pillar bundling resulted in a disordered surface geometry, the experimental and theoretical predictions of D_c all agree to within a factor of two or less, with no floating prefactors. Figures (b–f) reproduced with permission.⁵⁶ Copyright 2017, the American Chemical Society.

grow if its critical size exceeds $R_{\text{nuc}} \sim 1\text{--}10$ nm, as the volumetric energy decrease in chemical potential outweighs the surface energy increase due to expanding the new interface.⁶⁰ This critical embryo size is virtually always smaller than the gap within a unit cell of nanoroughness; combined with the lower energy barrier for heterogeneous nucleation at the bottom corners between the floor and roughness features, this means that dew tends to originate within the surface roughness.^{61–63}

Consider a minimal model case of a single nanodroplet nucleating inside of a unit cell of surface roughness comprised of four circular nanopillars.⁵⁶ It will proceed to diffusively grow within the unit cell until it becomes constrained by all four pillars, at which point, it will resemble a rectangular cuboid of width l and height l , where l is the center-to-center pitch between the pillars. Once uniformly constrained within the unit cell, the dew droplet must decide whether to grow sideways or upward (Figure 6b). The change in surface energy to grow sideways into an additional unit cell(s) by an incremental distance dx , $(\Delta E_{i,s})_{n \times n}$, versus the energy cost to grow upward by dz within the existing unit cell(s), $(\Delta E_{i,u})_{n \times n}$, can be expressed as the following ratio:

$$(E_i^*)_{n \times n} = \frac{(\Delta E_{i,s})_{n \times n}}{(\Delta E_{i,u})_{n \times n}} = \frac{(l-d)(1 - \cos \theta_A) - 2nl \cos \theta_A}{4(l-d) - n\pi d \cos \theta_A}, \quad (4)$$

where d is the diameter of each nanopillar, $\cos \theta_A$ is the (intrinsic) advancing contact angle on the smooth hydrophobic coating, and $n \times n$ is the number of unit cells that the droplet is currently constrained within ($n = 1$ to begin with, Figure 6b–d arbitrarily illustrates the case of $n = 2$). Lateral growth into a neighboring unit cell is expected when $(E_i^*)_{1 \times 1} < 1$, whereas upward growth toward the top of the surface roughness occurs for $(E_i^*)_{1 \times 1} > 1$. When the former case is true, it is assumed that 2×2 unit cells will get invaded, such that the width and height of the droplet are now $2l$ once the droplet becomes constrained again. Equation (4) is then checked, this time for $(E_i^*)_{2 \times 2}$, and this iterative process is repeated until either $(E_i^*)_{n \times n} > 1$ or the droplet reaches the tops of the pillars (i.e., $nl \approx h$, where h is the pillar height).

In the second stage of inflation, the droplet is now level with the pillar tops but needs to gain the ability to protrude outward into the ambient (Figure 6c). The energy ratio of sideways growth into additional unit cells versus upward inflation is given by⁵⁶:

$$(E_i^*)_{n \times n} = \frac{(\Delta E_{f,s})_{n \times n}}{(\Delta E_{f,u})_{n \times n}} = \frac{[(nl-d)(1 - \cos \theta_A) - 2nh \cos \theta_A]}{n\pi l}, \quad (5)$$

where the dx growth accounts for the wetting of the side walls and top faces of the nanopillars and the dz growth is approximated as a cylindrical disk of diameter l . When $(E_i^*)_{n \times n} > 1$ becomes true for the first time at a critical value of n_c , the droplet is able to inflate into the air with a pinned contact radius of $r_p \approx (1/2)n_c l$.

The third and final stage of inflation is for the droplet to increase its apparent contact angle to a sufficiently large value for jumping to be favorable (Figure 6d). A necessary criterion is that stage-three inflation thermodynamically favors the air-trapping Cassie state,

rather than the impaled Wenzel state, which is satisfied when $E^* = -1/(r \cos \theta_A) < 1$, where r is the roughness ratio of the surface.⁵⁹ Another key criterion is that the nucleation density must be much less than the feature density of the surface structure, such that each droplet has space to inflate without premature merging into other droplets, which would cause flooding.^{59,64} These dual criteria are usually satisfied for a robust nano-structured superhydrophobic surface, whereas they are virtually never satisfied for micro-only roughness.⁵⁹ The pinned contact radius r_p can be related to the droplet's apparent contact angle (θ^*) with the following geometric relation^{56,65,66}:

$$\theta^* = \cos^{-1} \left(\frac{r_p}{R} \right) + \frac{\pi}{2}, \quad (6)$$

where R is the radius of curvature of the droplet and the contact angles being considered are $\theta^* > 90^\circ$. On a robust superhydrophobic surface, a typical value for the advancing contact angle is $\theta_A \approx 160^\circ$. Once a droplet inflates to θ_A^* , its contact line will expand in a Cassie state beyond its pinned Wenzel neck. It follows that droplets need to inflate to θ_A^* for mature coalescence-induced jumping to be possible, to satisfy the criteria of large contact angles (to promote the downward expansion of the liquid bridge into the substrate) and low surface adhesion (of the expanded Cassie-state contact line, as opposed to the Wenzel state of the centralized impaled neck). The critical droplet diameter for jumping to occur can therefore be solved by rearranging Equation (6) and solving for $\theta_A^* \approx 160^\circ$ as a baseline for mature inflation⁵⁶:

$$D_c = \frac{2r_p}{\cos 70^\circ}. \quad (7)$$

On comparing this three-stage inflation model for D_c against the measured D_c values for six different nanopillar geometries, the agreement was excellent (Figure 6e,f).⁵⁶ When the model predicted that jumping was impossible for the two surfaces with very large nanopillars (i.e., $(E_i^*)_{n \times n} < 1$ even as $n \rightarrow \infty$), no real-life jumping was observed upon coalescence for any droplet size. For the four surfaces where the model predicted jumping was possible, the theoretical value of D_c matched the experimental one within a factor of two for three of the surfaces, without any fitting terms required. The one surface where the model and experiments were off by a larger factor of four, for the tallest and most slender nanopillars, was attributed to elastocapillary bundling occurring, which invalidated the model's assumption of vertical nanopillars. A later work by Vandadi et al. considered the more complex case of inflation within a hierarchical structure, where the Laplace-pressure-induced dewetting of droplets from the microcavities is possible.⁶⁷ In summary, it was conclusively demonstrated that the lower cutoff value for coalescence-induced jumping, R_c , is prescribed by the geometry of the underlying nanostructure, which governs the rate of mature inflation into a large-angle droplet with a Cassie-state contact line necessary for robust jumping. Finally, the limitations of inflation-based droplet coalescence can be bypassed entirely by depositing multiple droplets with a micropipette system and inducing coalescence via pushing or

droplet feeding.^{40,42} This approach simultaneously enables cleaner high-speed imaging, by virtue of not having neighboring condensate obscuring the focal plane.

15 VARIATIONS ON A THEME OF JUMPING

Variation #1: Enhanced condensation heat transfer

The phase-change heat transfer of dropwise condensation on a hydrophobic surface is typically five to seven times larger than filmwise condensation on a hydrophilic surface.^{68,69} This is because liquid condensate is a thermal barrier between the steam and the substrate; filmwise condensation is thick and continuous, which is unfavorable compared to the discontinuous coverage of dropwise condensation.⁷⁰ To sustain dropwise condensation, droplets must be continually removed from the surface as they grow to a critical size. Conventionally, this is achieved by gravitational removal using a vertical (or tubular) surface orientation.⁶⁹ The critical departure diameter scales to the capillary length: $D_c \sim L_c \sim (\gamma/\rho g)^{1/2}$, where g is the gravitational acceleration.^{71,72} For the dropwise condensation of water, $D_c \sim 1$ mm, which yields a heat transfer coefficient of $h \approx 100$ kW/m²K for $P_v \approx 100$ kPa steam when the noncondensable gases are removed.^{69,73,74}

A theoretical model of dropwise condensation revealed that the heat transfer coefficient increases monotonically with decreasing

D_c .⁷⁵ This model was confirmed by applying an external force to the condenser, such as vapor shear or centrifugation, to decrease the critical droplet size by an order of magnitude ($D_c \sim 0.1$ mm) and increase the heat transfer up to $h \approx 500$ kW/m²K for $P_v \approx 100$ kPa steam.^{76,77} The ability to continuously increase h with decreasing D_c has fascinating implications for condensation heat transfer, but the necessity of external forcing to lower D_c beneath the millimetric capillary length greatly hampers practical implementation.

It follows that the ability of jumping-droplet condensation to decrease D_c by two to three orders of magnitude compared to gravity, with no external forcing required, has attractive prospects for enhancing heat transfer. Indeed, this was the central conclusion of the original 2009 report,² in terms of forecasting prospective applications of the phenomenon. The first work to experimentally test this hypothesis was by Cheng et al.,⁷⁸ which surprisingly found inferior heat transfer coefficients for the jumping-droplet surface (i.e., two-tier superhydrophobic) compared to dropwise (smooth hydrophobic) or filmwise (smooth hydrophilic) condensation in a vapor chamber.

A landmark 2013 work by Miljkovic et al.⁶⁴ was the first to demonstrate that jumping-droplet condensation can be superior to dropwise condensation. Placing a superhydrophobic nanostructured tube comprised of silanized copper oxide within an evacuated vapor chamber (Figure 7a), they measured the flow rate and temperature increase of the cooling water inside the tube to extract the condensation heat transfer coefficient. A 30% higher heat transfer

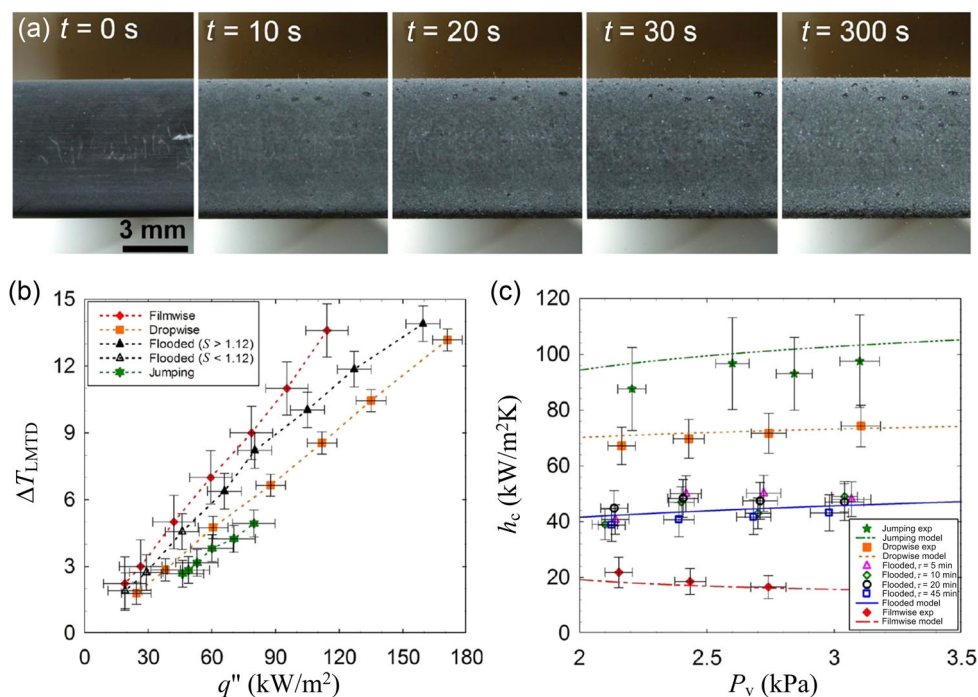


FIGURE 7 Superior phase-change heat transfer of jumping-droplet condensation. (a) Visualization of jumping-droplet condensation on a copper tube imparted with a superhydrophobic nanostructure. (b, c) At very low supersaturations, jumping-droplet condensation showed superior heat transfer compared to dropwise condensation. At modest supersaturations, flooding occurred, downgrading the heat transfer of the superhydrophobic tube to something between the dropwise and filmwise modes. Reproduced with permission.⁶⁴ Copyright 2013, the American Chemical Society.

coefficient was achieved by jumping-droplet condensation ($h \approx 100 \text{ kW/m}^2\text{K}$) compared to dropwise condensation ($h \approx 70 \text{ kW/m}^2\text{K}$). This was achieved by using low surface subcoolings ($\Delta T \lesssim 5^\circ\text{C}$) and corresponding steam pressures ($P_v \lesssim 3 \text{ kPa}$) to maintain the jumping mode of condensation (Figure 7b,c). At larger supersaturations, the jumping mode of condensation degraded to impaled flooding, resulting in inferior values of h compared to dropwise condensation in agreement with Jiang et al.⁷⁸

The flooding issue was largely mitigated by breakthrough work by Wen et al.^{79,80} By using very tall and closely packed nanowires (Figure 8a), the supersaturated water vapor was unable to deeply penetrate the interstices,⁷⁹ while the addition of 3D inter-connects minimized the structure's thermal resistance.⁸⁰ As a result, condensation preferentially formed on the upper portion of the nanowire network, preventing catastrophic flooding even at a much higher vapor pressure of $P_v \approx 60 \text{ kPa}$ and for ultra-large subcoolings of $\Delta T > 25^\circ\text{C}$ (Figure 8b). While the average departure size of droplets did increase at larger subcoolings due to partial flooding, resulting in a sharp decrease in h , it was always superior to the h values for dropwise condensation at an equivalent subcooling. In contrast, the performance of straight superhydrophobic nanowires degraded to match that of dropwise condensation by $\Delta T \approx 15^\circ\text{C}$. In terms of the magnitude of the heat transfer coefficient, even the highest value of $h \approx 71 \text{ kW/m}^2\text{K}$ is somewhat less than the $h \approx 100 \text{ kW/m}^2\text{K}$ measured by Miljkovic et al.,⁶⁴ so the primary impact of this report is in terms of the breadth of possible steam pressures and surface subcoolings due to the surface's antiflooding capacity.

A related work by the same group used alternating heights for the nanowire array, promoting preferential nucleation and Laplace-pressure-induced dewetting from the shorter bundles, to analogously delay flooding and enhance h .⁸¹ Seo et al. designed antiflooding superhydrophobic condensers by tailoring the heterogeneity of the surface coating and the length scale of the nanostructure, with a comparable finding of superior h measurements at large supersaturations.⁸² Very

recently, a record heat transfer coefficient of $h \approx 218 \text{ kW/m}^2\text{K}$ (at $\Delta T = 0.2^\circ\text{C}$) was achieved using a hierarchical superhydrophobic surface comprised of micropapillae structures and conformal nanograss, which promoted robust Laplace-induced dislodging from between the papillae for sustained jumping-droplet condensation.⁸³ The fundamental limits to jumping-droplet heat transfer have not yet been reached,^{84,85} which should continue to inspire enhanced surface architecture to minimize flooding and thermal resistance.

Variation #2: Characterizing droplet size distribution

The explicit measurements of h summarized above require an evacuated, custom-built vapor chamber, which is challenging and expensive to fabricate. For this reason, an increasingly common tactic is to indirectly calculate the heat transfer coefficient by plugging the steady-state droplet size distribution into an analytical thermal circuit model. Figure 9a shows the prototypical execution of this tactic for the case of dropwise condensation, where the droplet size distribution was varied by external forcing (using vapor shear or a centrifuge) and the experimental and analytical heat transfer coefficients show an excellent agreement.^{69,75–77} The complete analytical expression of this model is quite lengthy, but can be conceptually summarized as follows. First, the heat flux through a single droplet can be estimated as a function of its radius of curvature, contact angle, and the fixed temperatures of the saturated steam and condensing substrate.⁶⁹ Classically, the three sources of thermal resistance considered are the capillary and interfacial resistances across a droplet's surface and conduction across its bulk.⁶⁰ In the case of textured surfaces, additional corrective terms are required to account for the wetting state of the droplet and the thermal resistance of the air pockets and surface roughness.⁶⁶ A recent numerical work revealed the importance of convection as the droplets grow, such that the pure conduction model significantly underestimates the heat transfer.⁸⁷

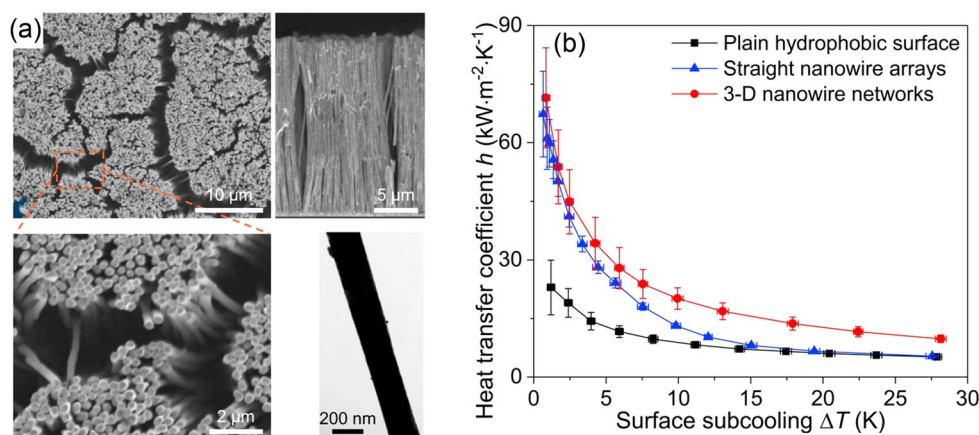


FIGURE 8 Mitigating the flooding issue for jumping-droplet condensation. (a) Scanning electron micrographs of a three-dimensional superhydrophobic nanowire network, which promoted preferential nucleation near the top of the close-packed bundles. (b) This prevented catastrophic flooding even at very large subcoolings, resulting in a more rugged demonstration of jumping-drop condensation for enhanced heat transfer. Reproduced with permission.⁸⁰ Copyright 2018, Elsevier.

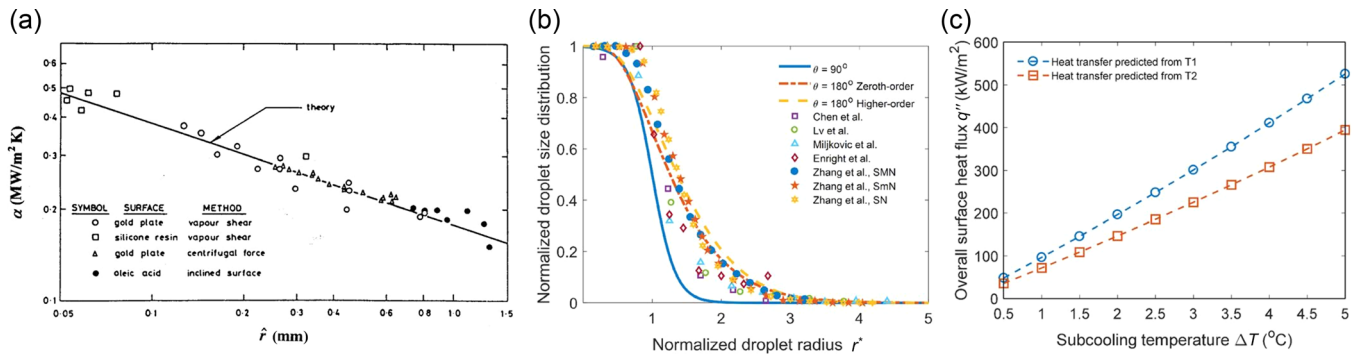


FIGURE 9 Effects of droplet size distribution on the condensation heat transfer coefficient. (a) Iconic graph from Rose,^{69,77} showing both experimentally⁷⁶ and theoretically⁷⁵ that the heat transfer coefficient increases monotonically with decreasing droplet departure size for dropwise condensation. Reproduced with permission.⁷⁷ Copyright 1988, Elsevier. (b) Example of an updated droplet size distribution model that is tailor-made for jumping-droplet condensation, overlaid with good agreement with experimentally measured distributions. (c) The updated population model (T1, blue circles) from (b) predicts a higher heat transfer coefficient compared to adapting the classic population balance theory developed for dropwise condensation (T2, orange squares). Figures (b, c) reproduced with permission.⁸⁶ Copyright 2019, AIP Publishing.

Second, a semiempirical population balance theory expresses the overall droplet size distribution as a function of the minimum possible embryo size ($D_{\min}$), the droplet size at which coalescence events first occur (D_{coal}), and the critical departure size (D_c), where either gravity or jumping removes droplets to define D_c .^{66,75} This droplet population model was originally crafted by Rose for the gravitational removal of dropwise condensation on a smooth surface;⁶⁹ in 2019, several updated size distribution models were crafted to account for the differing departure dynamics of jumping droplets.^{86,88,89} The updated droplet distribution model by Zhang et al.⁸⁶ is shown in Figure 9b.

Finally, once the size distribution has been estimated, the average heat flux through the entire surface is obtained by integrating the single-droplet heat flux equation over the entire droplet population, which can then be easily converted into the heat transfer coefficient using $h = q''/\Delta T$.^{66,69,77,87,90} Again looking at the updated size distribution theory for jumping droplets by Zhang et al.,⁸⁶ this results in a 33% increase in the predicted heat transfer coefficient compared to the classical population balance theory (Figure 9c). This is in better agreement with experimental measurements from Miljkovic et al.⁶⁴ While this theoretical approach to measuring heat transfer consumes dramatically less time than direct measurement, it is nonetheless cumbersome from an analytical standpoint, so the aspiring reader is encouraged to peruse the above references for the governing equations. There is still no firm consensus on when to account for convection or which population model to use.

Variation #3: Thermal diodes and vapor chambers

A thermal diode is more thermally conductive in one direction (forward mode) compared to the opposite direction (reverse mode). The effectiveness of a thermal diode is quantified by its dimensionless diodicity number⁹¹:

$$\eta \sim \frac{k_f - k_r}{k_r}, \quad (8)$$

where k_f and k_r are the effective thermal conductivities across the diode in the forward and reverse modes, respectively. The diodicity can be visualized graphically by plotting the applied heat flux versus the resulting temperature gradient across the device, where a steep slope is desired in the forward mode and a shallow slope in the reverse mode.

In theory, thermal diodes are highly desirable for the smart thermal management of electronics, aircraft, spacecraft, solar heaters, buildings, and many more applications. To date, the largest-scale use of thermal diodes has been in the Trans-Alaskan pipeline, where the vertical support pillars were comprised of thermosyphons to help preserve the permafrost layer.⁹¹ However real-life implementation of thermal diodes has been greatly constrained by the ultra-low diodicity of solid-state thermal diodes, the dependence of thermosyphons on gravity and a vertical orientation, and the 1D form factor of asymmetric heat pipes, which prevents scale-up to large 3D systems.⁹²

To work around these bottlenecks, Boreyko and Chen^{20,92} developed a jumping-droplet thermal diode by designing a planar vapor chamber with opposing superhydrophobic and superhydrophilic copper plates separated by an insulating gasket (Figure 10a). The superhydrophilic plate is typically comprised of a wick structure, for example, sintered copper, that is grafted onto the plate and holds the volume of degassed water charged into the chamber. When the wicked surface is heated with respect to the superhydrophobic one (forward mode), steam carries heat across the gap and condenses on the superhydrophobic surface; the self-propelled jumping motion returns the condensed droplets back to the evaporator, completing the circulation of working fluid with highly effective phase-change heat transfer. When the superhydrophilic surface is cooler (reverse mode), liquid water remains trapped in the wick and no phase-change heat transfer takes place; heat transfer is limited to ineffective conduction across the rubber gasket and vapor space. Performance is orientation independent, provided that the height of the vapor space is less than the maximum against-gravity jumping height of the

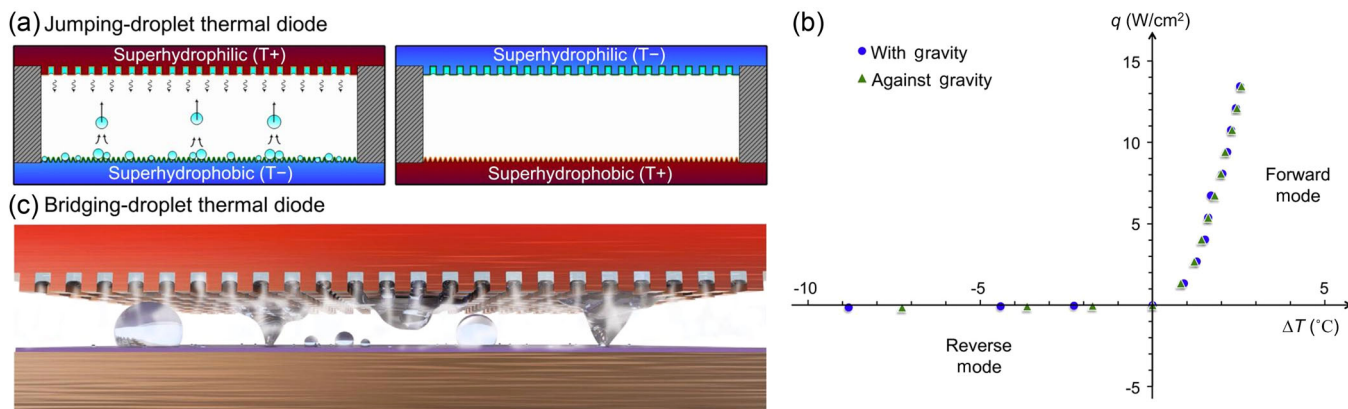


FIGURE 10 Phase-change thermal diode systems. (a) Schematic of a jumping-droplet thermal diode. In the forward mode of operation, dew self-propels from the condenser to return to the opposing wicked evaporator, enabling continuous phase-change heat transfer (left image). In the reverse mode, dew remains imprisoned on the colder wicked plate, resulting in dryout and insulation from the heated superhydrophobic surface. Reproduced with permission.⁹² Copyright 2011, AIP Publishing. (b) Diodicity map for a jumping-droplet thermal diode. The positive x-axis represents the forward mode, while the negative x-axis represents the reverse mode. The mismatch in thermal current is an analog to the electric diode used in computing. Reproduced with permission.²⁰ Copyright 2013, Elsevier. (c) The bridging-droplet thermal diode is the spiritual successor to the jumping-droplet thermal diode. By replacing the superhydrophobic condenser with a smooth condenser, jumping-droplet liquid return is abandoned and replaced with a simpler and more durable bridging-droplet return mechanism. Reproduced with permission.⁹³ Copyright 2020, John Wiley and Sons.

droplet (which is a function of both gravity and the air/vapor drag).^{20,46,94}

To date, the jumping-droplet thermal diode is the only material device that can achieve a gravity-independent diodicity of $\eta \sim 100$ across a scalable, two-dimensional form factor. When using a thin ($H_w = 0.5$ mm) wick on the evaporator side to minimize its conductive losses, and a high vapor temperature ($T_v \approx 80^\circ\text{C}$) to minimize the interfacial resistance of evaporation and condensation, an effective forward mode heat transfer coefficient of $h_f \approx 55$ kW/m²K was measured.²⁰ Note that for planar phase-change thermal diodes, we prefer to replace k_f and k_r with the corresponding effective heat transfer coefficients h_f and h_r , as the thermal resistance of the steam pathway is typically negligible and thus k_f can be nonphysically tuned by simply changing the length/height of the pipe/chamber. In contrast, the effective thermal conductivity in the reverse mode was only $k_r \approx 0.29$ W/m · K across a gap of $H = 2.6$ mm (including the wick),⁹² resulting in an effective heat transfer coefficient of $h_r \approx 0.11$ kW/m²K. This results in an estimated diodicity of almost exactly $\eta \approx 500$ (Figure 10b), which, to our knowledge, is the current record for thermal diodes. Follow-on work also demonstrated the ability of jumping-droplet vapor chambers to promote the rapid hot spot cooling of electronics.^{95,96}

Unfortunately, the superhydrophobic nanostructure needed to promote jumping-droplet liquid-return requires a hydrophobic monolayer coating (e.g., thiols and silanes); such coatings have been conclusively shown to break down under sustained exposure to steam.^{73,74,97} Therefore, the jumping-droplet thermal diode is not useful for real-life systems that need to operate continuously for months or even years without failure. For this reason, several recent reports independently developed bridging-droplet vapor chambers, where a smooth condenser replaces the fragile superhydrophobic

one.^{93,98–100} When the gap between plates is sufficiently small (typically $H_v \sim 10\text{--}100$ μm), dropwise condensate simply bridges across the gap to return to the wick upon reaching a critical size that scales with the gap height (Figure 10c).¹⁰¹ In addition to being durable, this approach is attractive because the engineered gap height directly defines any desired value of D_c , in contrast to dropwise or jumping-drop condensation where D_c is fixed to a single value for a given surface. As a result, the heat transfer can be tuned. A vertical hydrophobic condenser with an infinite gap height (i.e., no opposing wick) showed $h \approx 25$ kW/m²K by conventional gravitational removal, compared to $h \approx 35$ kW/m²K for bridging-droplet removal with $H_v \approx 600$ μm and $h \approx 90$ kW/m²K for $H_v \approx 180$ μm.¹⁰⁰ This latter value is comparable to jumping-droplet condensation, with no superhydrophobic structure required.

Variation #4: Electric-field enhanced jumping-droplet condensation

Miljkovic et al.¹⁰² keenly observed that when multiple jumping events occurred simultaneously in the field of view, the droplets repelled each other to alter their mid-flight trajectories (Figure 11a). This was attributed to the negative zeta potential of the superhydrophobic surface's conformational hydrophobic coating. As dew droplets nucleated and grew on the surface, OH[−] ions adsorbed to the coating, resulting in the formation of an electric double layer. The rapid removal of a jumping droplet therefore results in a net accumulation of H⁺ ions within the droplet. This was confirmed by placing an electrode near the condensing surface, which attracted the jumping droplets when negatively charged and repelled them when positively charged. The

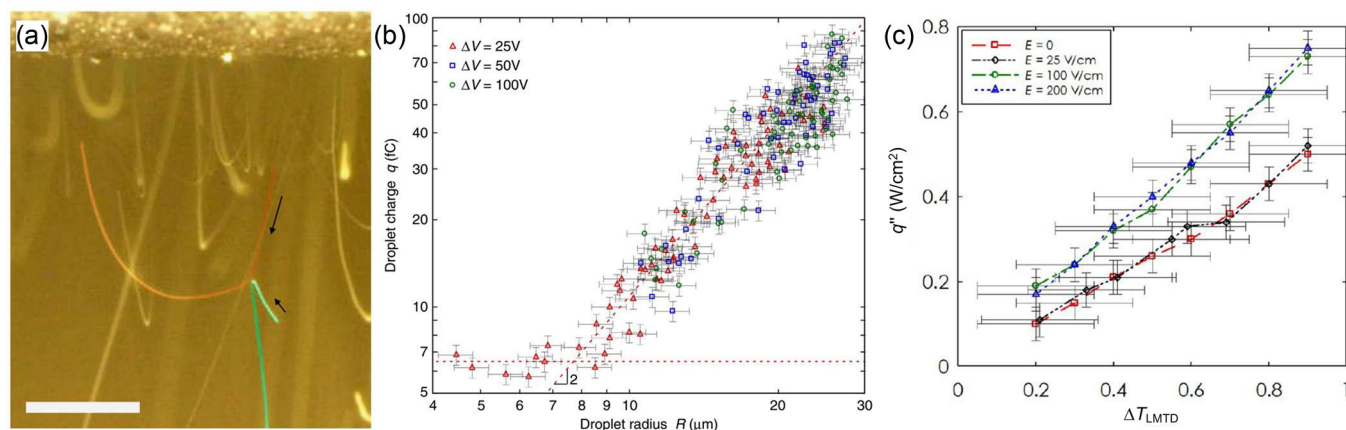


FIGURE 11 Electrostatic charging of jumping droplets. (a) Long-exposure photograph of the trajectories of jumping-droplet condensation. Two droplets repelled each other mid-flight (false-colored stream lines), implying electrostatic repulsion. (b) The charge of a jumping droplet was confirmed and measured by observing how an applied electric field modulated the jumping kinematics. Figures (a, b) reproduced with permission.¹⁰² Copyright 2013, Springer Nature. (c) By applying an electric field to a jumping-droplet condenser, the phase-change heat transfer was enhanced. Reproduced with permission.¹⁰³ Copyright 2013, the American Chemical Society.

charge of individual droplets could be extracted by applying a known electric field and measuring the droplet motions with high-speed imaging (Figure 11b); each droplet showed a charge of order $q \sim 10$ fC, with the magnitude increasing monotonically with droplet size.

Electrostatic charging of jumping droplets can be exploited in two opposite manners. By actively applying a negative charge to a cylindrical copper mesh that surrounds a condenser tube, the heat transfer coefficient of jumping-droplet condensation can be boosted by at least 50% (Figure 11c).^{103,104} Electric-field-enhanced jumping droplets were also used to roughly double the diodicity of a jumping-droplet thermal diode vapor chamber.¹⁰⁵ Conversely, when droplets jump from a superhydrophobic copper condenser to an opposing hydrophilic copper surface, an electrostatic potential is formed, which can be used for energy harvesting.¹⁰⁶ A power density of ≈ 15 pW/cm² was demonstrated experimentally and it was speculated that densities as high as ≥ 1 μ W/cm² are possible at higher condensation rates.

Variation #5: Convective condensation

In many real-life applications, the vapor resulting in condensation is convective in nature, either flowing internally within tubes or externally over tubes or plates/fins. In 2013, Torresin et al.⁹⁷ observed condensation on a nanostructured superhydrophobic surface that resulted from saturated steam flowing at a high vapor temperature ($T_v = 110^\circ\text{C}$), pressure ($P_v = 1.43$ atm), and shear rate ($v_v = 6, 12$, or 18 m/s). It was found that the heat transfer coefficient increased with v_v . However, the extreme environmental conditions resulted in dropwise condensation, rather than jumping-droplet condensation, analogous to previous work where jumping ceased at large supersaturations. Further, by the sixth day of flow condensation, even the dropwise mode was lost, resulting in filmwise condensation that was inferior to a smooth and uncoated oxidized copper tube.⁶⁴

A pair of theoretical studies by Birbarah and Miljkovic modeled the performance of convective jumping-droplet condensation both internally (in a pipe)⁹⁴ and externally (on a flat plate).¹⁰⁷ For internal vapor flow within a horizontal pipe, the trajectory of jumping droplets is modified by a drag force and also by the Saffman lift force⁹⁴ (an excellent conceptual introduction to the latter was written by Howard Stone¹⁰⁸). In turn, the modified trajectories of jumping droplets affect the rate of multihop chain reactions that occur downstream of a coalescence event. The pressure drop across the pipe was modeled as a function of the work done on the vapor by the pipe walls and droplets, with the simplification that flooding and/or stuck droplets were neglected in the model. For the theoretical model of jumping droplets on an external plate, the effects of the boundary layer and plate inclination on droplet trajectories were considered.¹⁰⁷

Internal convective jumping-droplet condensation was realized experimentally by Antao et al. in 2020.¹⁰⁹ Copper tubes were used as the material, and all three modes of condensation (filmwise, dropwise, and jumping-droplet) were compared by varying the surface wettability. For the superhydrophobic tube, a transition from jumping to flooding occurred at a log mean temperature difference of $\Delta T_{LM} \approx 2^\circ\text{C}$ between the water vapor and the tube wall. As a result, the heat transfer coefficient of the superhydrophobic tube was superior at low heat fluxes ($h \approx 68$ kW/m²·K at $q'' \approx 50$ W/cm²), but inferior to dropwise and comparable to filmwise at large heat fluxes ($q'' > 50$ W/cm²). This work also confirmed the intuition that all three modes of condensation should show increasing values of h with increasing inlet vapor velocity, which ranged from 20 to 40 m/s.

Variation #6: Momentum model for jumping

The conventional jumping-droplet energy model presumes that the released surface energy is entirely transferred to translational kinetic

energy and over-predicts the jumping velocity by a factor of five. An alternative analytical model, based on conservation of momentum, was developed by Mouterde et al.¹¹⁰ to resolve this discrepancy. This momentum approach allows for the modeling of dissimilar droplet sizes, where one droplet is of radius R but the second droplet is of smaller radius r (Figure 12a). Microfabrication was used to install a force sensor beneath jumping droplets to validate the model with experimental measurements of the jumping force (Figure 12b).

Beginning with the simplified case of two same-sized droplets ($r = R$), the outer edge of each droplet retracts during coalescence by a distance of $\approx R$. For the typical jumping-droplet scenario where $Oh \ll 1$, this retraction is capillary-inertial with a time scale of $\tau_r \approx 2\sqrt{\rho R^3/\gamma}$, where the prefactor of 2 is from a best fit to the experimental data. The retraction velocity is therefore $v_r \approx R/\tau_r$.

By conservation of momentum, $mv_r = 2mv_j$, where m is the mass of each droplet pre-coalescence. The left-hand side of the equation is not doubled because roughly half of each droplet's mass is retracted from the droplet's edge during coalescence. Putting everything together, this yields:

$$v_j \approx \frac{v_{ci}}{4} = \frac{1}{4} \left(\frac{\gamma}{\rho R} \right)^{1/2}. \quad (9)$$

The coefficient of $1/4$ to relate the jumping velocity with the capillary-inertial velocity is remarkably close to the experimentally measured value of $v^* \approx 0.22$, with an error of only about 10% (Figure 12c).⁴¹ This innovative approach also matched the experimental data quite well over a wide range of size mismatches between R and r , without requiring an adjustable parameter or numerical simulations (Figure 12d).^{41,110} For smaller micro-droplets approaching the critical size for jumping, the model over-predicts the real-life jumping speed.⁴¹ The authors address

this by modifying the momentum expression, adding a function of Oh to account for viscous effects, but we suspect that immature droplet inflation is the more likely culprit for the deviation as discussed in the previous section.

Variation #7: Enhanced jumping with micro/macro-structures

Recall that the physical mechanism for out-of-plane jumping is the underlying substrate breaking the symmetry of the radially expanding liquid bridge during coalescence.^{2,21,28} It follows that, by enhancing symmetry breaking by placing an underlying microstructure beneath coalescing droplets, the efficiency of the jumping process can be enhanced. Compared to the poor efficiency of $v^* \approx 0.2$ and $\eta \approx 4\%$ on homogeneous superhydrophobic nanostructures, in 2016, an efficiency of $v^* \approx 0.53$ and $\eta \approx 22\%$ was reported by incorporating a macrotexture (in the shape of a triangular prism) into the superhydrophobic surface.¹¹¹ When this protrusion is directly beneath the droplets before coalescence, it boosts the extent of symmetry breaking to enhance the hydrodynamics of converting surface energy into translational kinetic energy.

In 2018, a comparable efficiency of $\eta \approx 19\%$ was achieved by using a more slender triangular prism (Figure 13a).¹¹² This report featured an elegant scaling model to simply rationalize the boost in efficiency. Recognizing that additional surface energy is stored as the merging droplet wraps around the ridge, an extra term is added to the energy balance:

$$C_1 \gamma R^2 + C_2 \gamma R h_{\text{ridge}} \approx \rho R^3 v_j^2, \quad (10)$$

where the second term accounts for the energy released as the droplet departs from the ridge, h_{ridge} is the ridge height, and C_1 and C_2

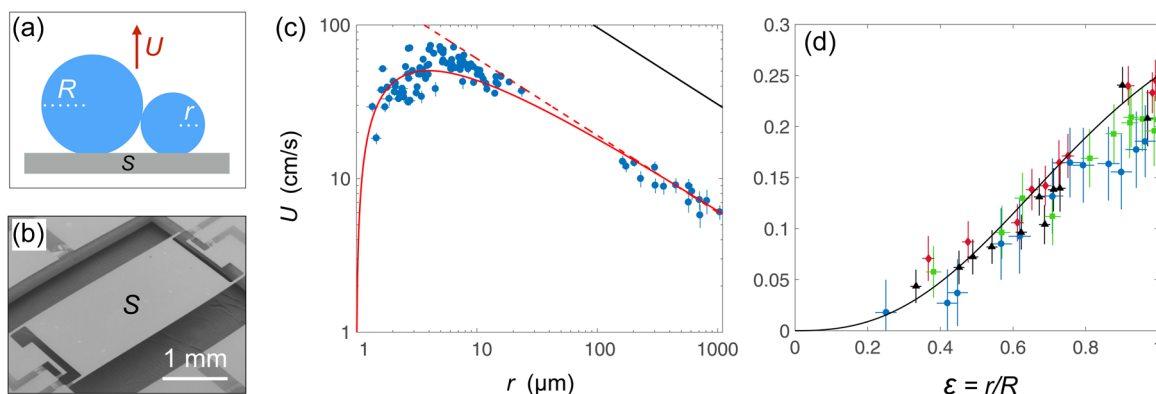


FIGURE 12 Conservation of momentum model for jumping droplets. (a) The model allows for dissimilar droplet sizes. (b) A microelectromechanical sensor was designed for direct force measurements during droplet jumping. Figures (a, b) reproduced with permission.¹¹⁰ Copyright 2017, the American Physical Society. (c) The momentum conservation model (solid red line) was in good agreement with the experimental data (blue circles) for same-sized coalescence events. The model captures the downward trend at smaller values of r by adding an Oh term to the model, which we argue might be an ad hoc approach to account for inflation effects. Regardless, the model approximates the proper value of $v^* \approx 0.22$ (red dashed line) for larger droplet sizes, which is well below the idealized capillary-inertial velocity (solid black line). Reproduced with permission.⁴¹ Copyright 2019, the American Physical Society. (d) The model is excellent at capturing the effects of size mismatch without any adjustable parameters. Reproduced with permission.¹¹⁰ Copyright 2017, the American Physical Society.

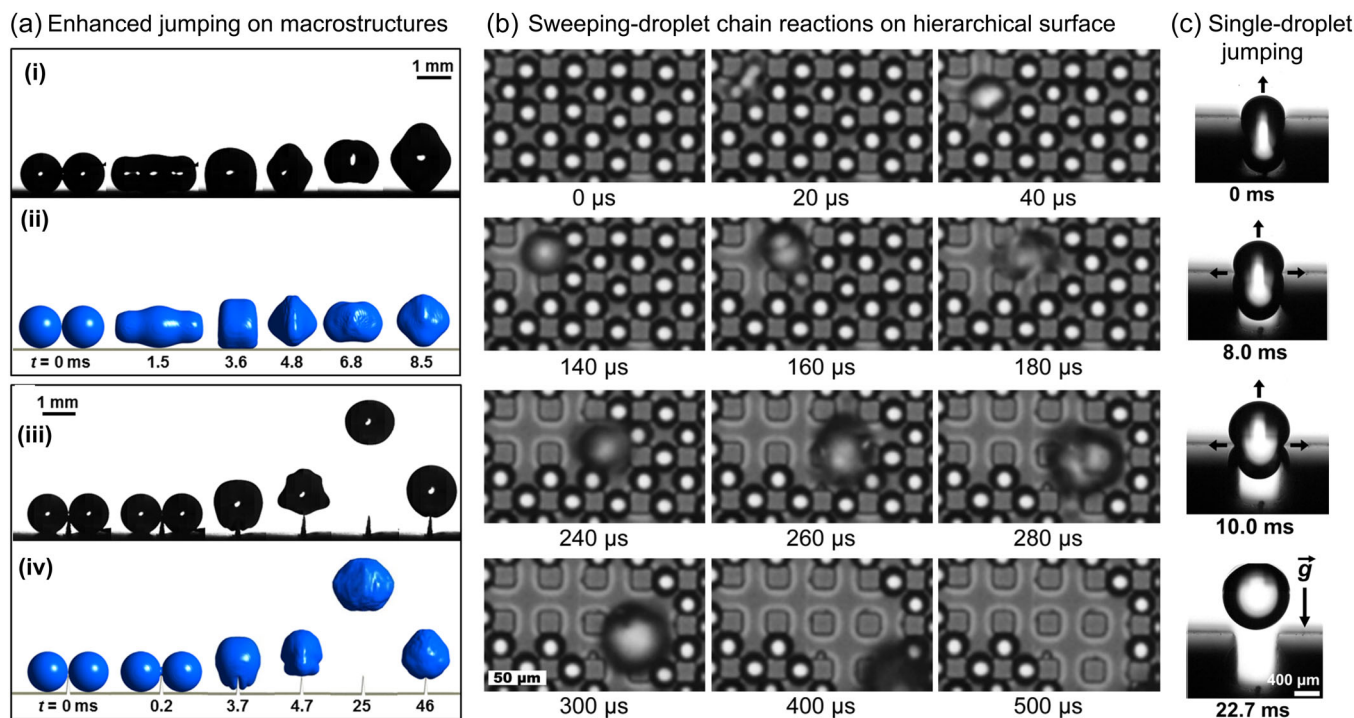


FIGURE 13 Three alternatives to conventional droplet jumping. (a) Compared to the low v^* values of conventional droplet jumping (i, ii), coalescence over a macrostructure can dramatically boost v^* (iii, iv). Reproduced with permission.¹¹² Copyright 2018, AAAS. (b) By nucleating dew along the side walls of a microstructure coated with a conformal nanostructure, self-propulsion is now lateral rather than out-of-plane. This results in sweeping-droplet condensation, eliciting desirable chain reactions analogous to gravitational sweeping for dropwise condensation. Reproduced with permission.⁶² Copyright 2015, AIP Publishing. (c) Single-droplet jumping is possible when a droplet is squeezed within a micro-cavity. This bypasses the need for coalescence entirely and can even result in values of v^* approaching unity. Reproduced with permission.¹¹³ Copyright 2020, the American Chemical Society.

are prefactors that account for geometry and partial energy conversion. Defining the jumping Weber number as $We_j = v^*{}^2 = \rho R V_j^2 / \gamma$, Equation (10) can be nondimensionalized as:

$$C_1 + C_2 h^* \sim We_j, \quad (11)$$

where $h^* = h_{\text{ridge}}/R$. Given that $\eta \propto We_j$, Equation (11) implies that the jumping efficiency increases linearly with h^* , which was confirmed by varying h^* experimentally.

A new record efficiency of $v^* \approx 0.61$ and $\eta \approx 30\%$ was reported in 2020 using an asymmetric V-groove.⁴³ Before coalescence, one droplet was placed at the corner above the groove, while the second droplet was placed within the groove. Months later, the record was broken again by using symmetric micro-grooves, achieving $v^* \approx 0.74$ and $\eta \approx 46\%$.¹¹⁴ Figure 14 replicates the centerpiece graph of v_j versus R from the main theme, but this time, the data points represent these cases of macrostructure-enhanced jumping. The data from the triangular prism,¹¹² asymmetric V-groove,⁴³ and symmetric micro-grooves¹¹⁴ are all well above the inefficient $v^* \approx 0.17$ – 0.26 range for a uniform superhydrophobic surface, but of course still below the idealized capillary-inertial velocity curve. The case of the asymmetric micro-grooves resulted in a diagonal jumping motion, as opposed to out-of-plane with respect to the floor, which is an excellent transition to the related topic of directional jumping.

The out-of-plane directionality of conventional droplet jumping is due to the substrate breaking the symmetry of the expanding liquid bridge. While the initial trajectory of most jumping droplets is not perfectly out-of-plane,² the launching angle is semirandom due to its complex dependence on the mismatch in size (and Laplace pressure) of the merging droplets or, for groups of three droplets or more, their geometric configuration.^{50,51,115,116} For chain reactions, where a launched droplet returns to the surface for a subsequent jump, the jumping angle additionally depends on the angle and speed with which the returning droplet lands and collides with sessile droplets.^{52,117}

But what if a micro- or macro-texture, such as a side wall, additionally breaks the symmetry of the liquid bridge? The result should be a diagonal launching even for same-sized droplets, at more dramatic angles that are repeatable and predictable. When launching at a large angle with respect to the out-of-plane axis, jumping droplets can easily collide with neighboring dew droplets, resulting in massive chain reactions of droplet sweeping. In this respect, sweeping via directional droplet jumping is reminiscent of the chain reactions driven by gravity-induced sweeping during dropwise condensation on a vertical wall, but at much smaller length scales. A single jumping event triggering the rapid sweeping removal of many micro-droplets is highly attractive in the context of condensation heat transfer or promoting antifogging. In contrast,

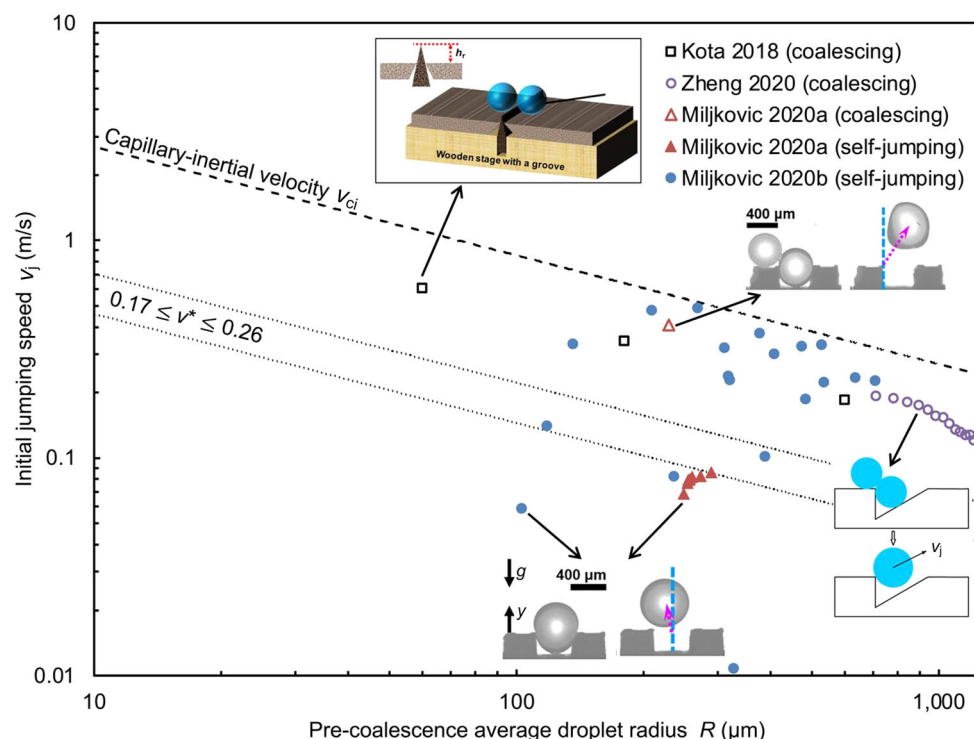


FIGURE 14 Composite results for enhanced jumping via coalescence on macrostructures (open symbols) or single-droplet jumping via Laplace pressure mismatch (filled symbols). The jumping efficiency can vary widely, but values of v^* tend to lie above the 0.17–0.26 band occupied by conventional droplet jumping and can even approach the capillary-inertial limit (dashed black line) in some cases. Insets denote the specific micro/macrot textures used: coalescence over a triangular prism,¹¹² an asymmetric V-groove,⁴³ or a micro-groove,¹¹⁴ or self-jumping from microgrooves.^{113,114}

with conventional jumping, the micro-droplets are removed by discrete coalescence events, without nearly as many chain reactions. Further, the chain reactions for conventional jumping are gravity-driven and require a horizontal (or slightly tilted) surface orientation, whereas directional sweeping is capillary-inertial and orientation-independent.

To our knowledge, the first mention of droplet sweeping is from Chen et al.⁶³ Using a hierarchical copper superhydrophobic surface, they observe that “a majority of coalescence-induced droplet departure events take place in between adjacent microposts and multiple participating droplets are involved,” resulting in the “sweeping of mobile droplets off the surface.” However, no high-speed imaging of the sweeping was captured and the effect was not discussed further. (The original 2009 report also used a hierarchical surface, but the micropillars were much smaller, such that droplets inflated atop the micropillars rather than between them.²) Also in 2015, Qu et al.⁶² were the first to capture high-speed imaging of self-propelled sweeping, using a hierarchical surface comprised of silicon micropillars conformally coated with an aluminum hydroxide nanostructure. Triggered by a single coalescence event, 15 micro-droplets were swept away from the field of view of the microscope within only 0.5 ms (Figure 13b). Numerical simulations confirmed that the mechanism was the additional symmetry breaking provided by the two adjoining side walls of the micropillar. Follow-up reports demonstrated directed jumping and subsequent droplet sweeping using triangular micro-arrays,¹¹⁸ hierarchical nanowires,⁸¹ zigzagging microstructures coated with nanoglass,¹¹⁹ V-shaped

macrogrooves,¹²⁰ cactus-inspired wedge-shaped bumps,¹²¹ and the aforementioned asymmetric V-groove.⁴³

Variation #8: Single-droplet jumping

A practical limitation to jumping-droplet condensation is its dependence on coalescence events. This is problematic for two reasons in particular. First, when the nucleation density is only slightly weaker than the packing density of the surface structures, embryos growing within the texture merge together in an impaled state before their contact lines are able to inflate upward into the necessary Cassie state.⁵⁹ Second, coalescence events are often between droplets featuring large mismatches in size, in which case no jumping occurs upon coalescence because the liquid bridge no longer impacts the substrate appreciably.^{45,51} If droplets fail to launch after multiple coalescence events, they become so large that even same-sized coalescence can no longer propel them from the surface.⁹⁰

An attractive solution to these issues with coalescence-induced jumping is to instead induce single-droplet jumping events. This can be accomplished by using superhydrophobic surfaces that cause an asymmetry in a droplet's Laplace pressure during its growth and confinement within the structure. A prelude to this concept was provided by Lv et al.¹²² in 2015, who showed that Laplace-pressure-induced squeezing could cause spontaneous Wenzel-to-Cassie transitions on hierarchical superhydrophobic surfaces. For example,

in 2016, Aili et al. showed that by overlaying a nanotextured copper micro-mesh on a nanotextured copper plate, isolated dew droplets could self-jump from the surface as they began to bulge out of the holes in the mesh.¹²³

In 2020, Yan et al.¹¹³ showed Laplace-driven single-droplet jumping from a monolithic surface, comprised of nanotextured micro-grooves (Figure 13c). Further, by using asymmetric grooves where the vertical wall was taller on one side, directional single-droplet jumping was achieved. By optimizing the groove size and initial droplet volume, an impressive jumping efficiency of $v^* \approx 0.95$ and $\eta \approx 60\%$ was achieved, well above what is possible with coalescence-induced jumping even with the assistance of macrotextures.^{113,124} The hydrodynamic efficiency of single-droplet jumping varied widely, as shown in Figure 14. The earlier report that was more focused on coalescence-induced jumping featured v^* values comparable with a uniform superhydrophobic surface,¹¹⁴ while the follow-up report focused on single-droplet jumping approached the capillary-inertial performance ceiling for the optimum choice of groove width and droplet size.¹¹³

Variation #9: Droplet jumping on biphilic surfaces

While nonflooded jumping-droplet condensation shows superior droplet shedding and phase-change heat transfer compared to dropwise and filmwise condensation,⁶⁴ there are still aspects of superhydrophobic condensers that are suboptimal. The energy barrier for nucleating a fresh liquid embryo is larger compared to a hydrophilic surface, which negatively impacts the nucleation density.⁶⁰ The conduction resistance of large-angle and air-trapping Cassie droplets is large compared to drops of equivalent volume on smooth hydrophobic surfaces.⁶⁶ Another issue is that coalescence events on homogeneous superhydrophobic surfaces frequently suffer from droplet size mismatch, greatly reducing the odds of jumping.^{45,58}

To obtain the best of both worlds, hybrid surface structures have been engineered that contain both hydrophilic (or smooth hydrophobic) features, to enhance and/or spatially control the nucleation

density and increase growth rates, and superhydrophobic features, to promote droplet jumping. To our knowledge, this concept of biphilic jumping-droplet condensation was first introduced by Chen et al.⁶¹ Their surface was comprised of an array of micropyramids coated with nanograss, where the base of the micropyramids was smooth without any nanograss (Figure 15a). Droplets preferentially nucleated about the smooth side walls of the micropyramid bases, followed by growth and suspension atop the pyramids for coalescence-induced jumping. This approach resulted in a 65% increase in the droplet number density and a 450% increase in the total volume of droplet self-removal during condensation.

The results reported by Chen et al.⁶¹ strongly indicate that patterned surfaces should enhance phase-change heat transfer, but h was not measured. Hou et al.¹⁸ engineered wettability patterns onto a surface by protruding a dilute array of hydrophilic microposts above a field of superhydrophobic nanograss (Figure 15b). A more extreme version of the smooth hydrophobic side walls of the micropyramid surface, the dilute hydrophilic microposts enhanced the nucleation and growth of droplets, while their uniform spacing promoted same-sized coalescence events. This time, the heat transfer was measured, finding a 60% improvement in h compared to dropwise condensation (i.e., double the improvement reported by Miljkovic et al.⁶⁴ discussed in the first variation above). However, there is the sharp caveat that all h values were orders of magnitude lower, due to the noncondensable gases not being purged. Again, to avoid flooding issues, low partial pressures were used for the water vapor within the chamber ($P_v \lesssim 3$ kPa).

A similarly patterned biphilic surface, by Ölçeroğlu and McCarthy,¹²⁵ further demonstrated that the spatial control of same-sized coalescence events can prevent flooding at a moderate supersaturation of $S = 1.54$, whereas a uniform superhydrophobic surface was heavily flooded. (Supersaturation is calculated using $S = P_v/P_{\text{sat}}$, where P_v is the partial pressure of water vapor in the ambient and P_{sat} is the saturation pressure corresponding to the temperature of the subcooled condenser.) A numerical study by Hoque et al.⁵³ returned to the concept of using smooth hydrophobic spots, rather than hydrophilic spots, to minimize the adverse effects of droplet pinning. Their central hypothesis was to use a biphilic

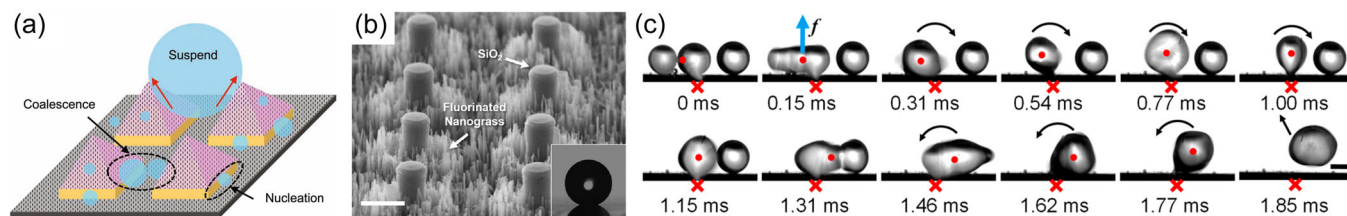


FIGURE 15 Jumping-droplet condensation on biphilic surfaces. (a) A hierarchical superhydrophobic surface featuring smooth, hydrophobic side walls for the base of the micro-features. Condensation preferentially nucleated and grew from the smooth side walls for enhanced nucleation density and jumping-droplet removal. Reproduced with permission.⁶¹ Copyright 2011, John Wiley and Sons. (b) Another approach to a biphilic surface is to use dilute hydrophilic features (SiO_2 micro-posts) for enhanced spatial control and growth of nucleating droplets. Reproduced with permission.¹⁸ Copyright 2015, the American Chemical Society. (c) Biphilic surfaces can modify the hydrodynamics of jumping. Here, a droplet swings around a droplet pinned to a hydrophilic feature during coalescence, resulting in a collision and directional jump with a third droplet. Reproduced with permission.⁴² Copyright 2019, the American Chemical Society.

surface to promote a contrast in droplet growth rates, with faster growth occurring on the hydrophobic spots while using the superhydrophobic background for jumping-induced departure. Their model predicts an order-of-magnitude increase in the condensation heat transfer coefficient compared to a homogeneous superhydrophobic surface.

In addition to increasing nucleation density and the self-removal volume, the hydrodynamics of droplet jumping are also modified on biphilic surfaces. Using deposited microdroplets on a hierarchical biphilic surface featuring hydrophilic hilltops, Yan et al.⁴² observed that the jumping velocities of same-sized droplets are lower with more scatter compared to uniform superhydrophobic surfaces. This is due to the greatly increased work of adhesion for any droplet detaching from a hydrophilic patch. When only one of the two droplets was pinned to a hydrophilic feature during coalescence, original modes of directional jumping were observed. For example, the unpinned droplet could swing around the top of the pinned droplet during coalescence, in some cases even coalescing into an additional droplet, which then resulted in a lateral jump (Figure 15c).

With alternative approaches to biphilic surface design, it is possible to enhance the jumping velocity rather than diminish it. Zhu et al.¹²⁶ observed enhanced jumping velocities on a biphilic surface, rather than lower velocities, by seeding nanometric hydrophilic sites atop superhydrophobic nanograss. As a result, droplets tended to nucleate atop the surface structure with minimal pinning on the ultra-small hydrophilic sites, rather than inflating from within the surface structure with higher-adhesion Wenzel necks as previously discussed. A follow-up work optimized this approach, resulting in a 43% increase in condensation heat flux and a 61% increase in water collection flux compared to a homogeneous superhydrophobic surface.¹²⁷ When placing the hydrophilic spot on the bottom of the microstructure, rather than the top, Stando et al.¹²⁸ numerically

demonstrated a fun droplet “slingshot” effect. By using truncated cones as the microstructure, droplets show an upward Laplace-pressure driven force while simultaneously being pinned to the floor. This serves to elongate the stretching before take-off, culminating in a more dramatic, slingshot-like, ejection from the surface.

Variation #10: Universal jumping on nanocones

The nucleation of a minimal liquid embryo tends to occur within the surface roughness, such that even a successfully inflated Cassie droplet features a centralized Wenzel neck.^{56,90,129} This raises an important question: during jumping-droplet condensation, what percentage of the unit cells of roughness is impaled with water due to the presence of Wenzel necks (or flooded droplets)? This question is difficult to answer, as nanoscale roughness cannot be resolved optically and, even for microscale roughness, it is difficult to look underneath a dew droplet.

In 2017, Mouterde et al.⁵⁸ devised a simple and elegant solution to this issue. First, they nanofabricated ordered arrays of cylindrical nanopillars versus nanocones by using electron-beam lithography and block-copolymer self-assembly (Figure 16a). Each substrate was then inclined at a known angle and a large, warm droplet was held in place attached to a syringe pump needle. The increased temperature of the droplet relative to the room temperature substrate, ΔT , induced condensation within the underlying nanotexture. By continually feeding the droplet from a needle attached to the syringe pump, the droplet eventually detached from the needle at a critical mass to slide down the superhydrophobic surface. The lateral adhesion force of the superhydrophobic surface is then extracted from:

$$F = mg \sin \alpha - \pi d \gamma, \quad (12)$$

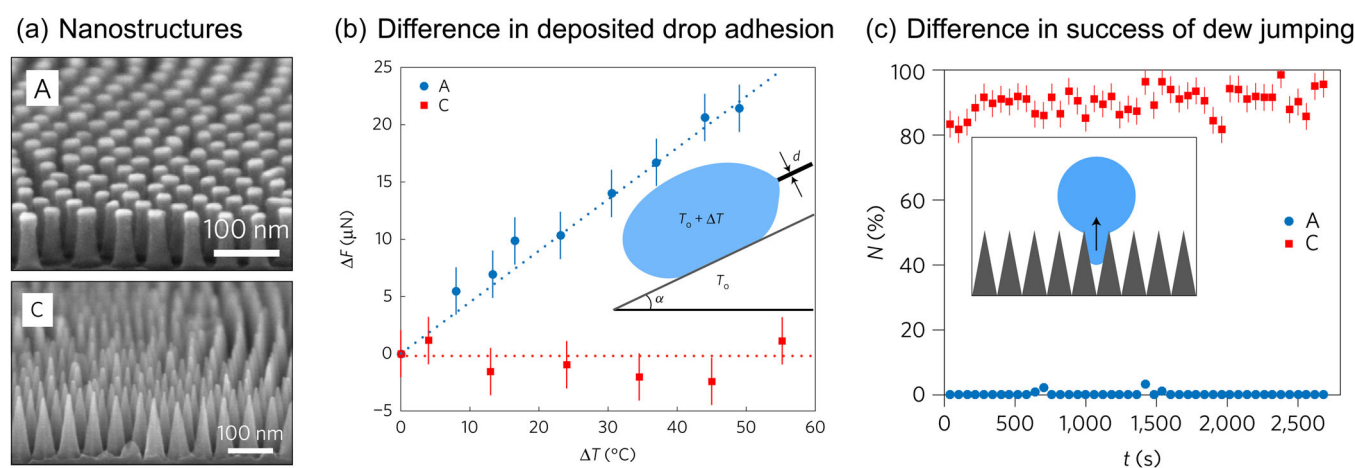


FIGURE 16 Universal jumping on superhydrophobic nanocones. (a) Cylindrical nanopillars were compared against nanocones. (b) The adhesion force of a warm Cassie droplet overlying condensation steadily increased with droplet temperature for the nanopillars, but remained negligible for nanocones. The latter result implies a spontaneous dewetting mechanism for embryos growing between nanocones. (c) Probability of jumping upon a coalescence event for both surfaces during condensation growth. The spontaneous dewetting and rapid inflation of dew on nanocones resulted in near-universal jumping success upon coalescence events (red squares), compared to the nanopillars where jumping rarely occurred (blue circles). Reproduced with permission.⁵⁸ Copyright 2017, Springer Nature.

where m is the critical mass of the detaching droplet, α is the surface's angle of inclination, and d is the needle diameter. Recognizing that a baseline adhesion, F_0 , exists even in the absence of condensation due to contact angle hysteresis, the condensation-induced adhesion force is deduced using $\Delta F = F(\Delta T) - F_0$. When using densely packed cylindrical nanopillars, the deduced values of ΔF only increased weakly with ΔT . In contrast, dilute cylindrical nanopillars with a pillar density of only 10% showed a sharp curve for ΔF versus ΔT (i.e., much larger values of m). This was rationalized by comparing the nucleation density of condensation to the packing density of the pillars, quite analogous to the prior theory for flooding versus Cassie inflation provided by Enright et al.⁵⁹ in 2012.

Thus far, Mouterde et al.⁵⁸ are simply providing a clever method for deductively measuring the extent of flooding during condensation. The report then takes a surprising turn, by replacing the nanopillars with nanocones to discover that ΔF vanishes entirely for any value of ΔT (Figure 16b). Two factors were attributed to the stellar performance of the nanocones: the tapering of the interstitial gaps hinders condensation at the bottom of the structure (as also shown by Wen et al.¹³⁰), and the mismatch in droplet curvature can induce a Laplace-driven expulsion out of the texture. This latter factor was confirmed by removing the deposited droplet and simply observing jumping-droplet condensation. Whereas most coalescence events did not result in jumping for the cylindrical nanopillars, $\approx 90\%$ of all coalescing droplets jumped from the nanocones (Figure 16c). This increased to $\approx 99\%$ when only considering same-size droplets. A follow-up study found that the remarkable antifogging ability of superhydrophobic nanocones is universal across a wide range of cone sizes, apex angles, and cone shapes.¹³¹ The critical radius for coalescence-induced jumping was a mere $R_c \approx 1.5 \mu\text{m}$, approaching (but not yet reaching) the nanoscale.

Variation #11: Nanodroplet jumping

Nanodroplet jumping is possible in theory, as viscous effects do not dominate capillary-inertial ones (i.e., $Oh_c \geq 1$) for water until a critical droplet size of $R_c \lesssim 10 \text{ nm}$.² However, from a purely practical standpoint, demonstrating coalescence-induced jumping with nanodroplets is virtually impossible. Jumping-droplet condensation on superhydrophobic surfaces is generally confined to microscopic droplet sizes of $D_i \geq 1 \mu\text{m}$, as nanodroplets have not fully inflated as previously discussed.^{56–58} While inflation is irrelevant for the context of condensation or fog on a smooth hydrophobic fiber, nonetheless, large droplet-to-fiber ratios are required that similarly require micro-droplets.¹³² Using a Leidenfrost stage avoids these issues, but Leidenfrost micro-droplets spontaneously take off into the air or explode as they evaporate to a critical size.^{133,134} These collective practical limits: inflation on superhydrophobic surfaces, requiring large droplet-to-fiber ratios for fibers, and the imminent expulsion of ultra-small Leidenfrost droplets, have prevented any experimental observations of jumping nanodroplets. Even if nanodroplet jumping were possible in real life, it would be nontrivial to capture optically due to the diffraction limit.

For these reasons, to date, numerical simulations have been the exclusive domain for investigating nanodroplet jumping. To our knowledge, the first molecular dynamics (MD) simulation of coalescence-induced jumping nanodroplets was performed by Liang and Kebinski (Figure 17a).¹³⁵ Their model system involved same-sized liquid Ar nanodroplets coalescing on a smooth gold substrate whose Ar–Au interaction was adjusted to mimic superhydrophobicity. They found that the expansion of the liquid bridge was still capillary-inertial in nature for nanodroplets and that jumping successfully occurred over the entire range of droplet sizes ($D = 23–53 \text{ nm}$) and Ohnesorge numbers ($Oh = 0.36–0.55$). The

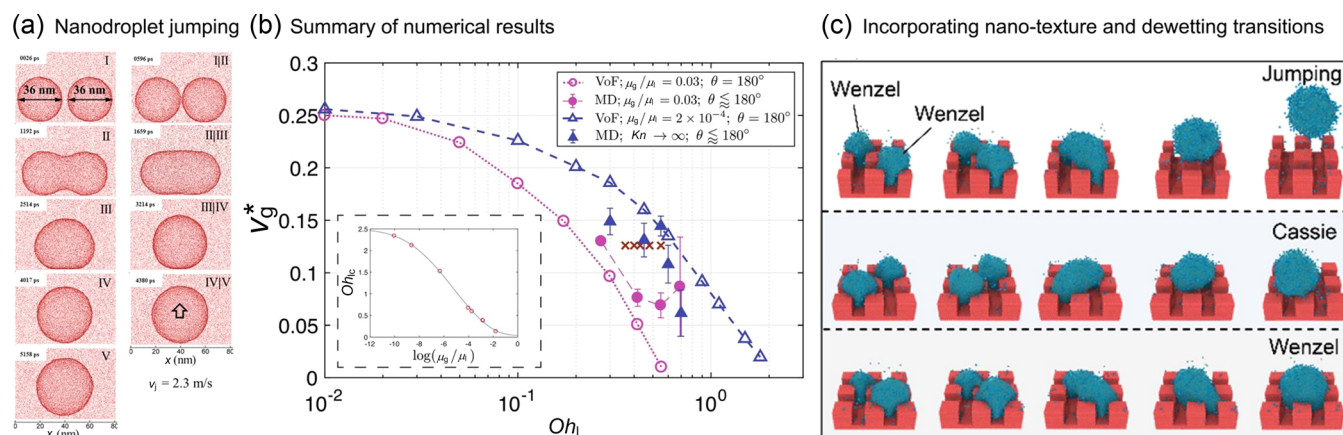


FIGURE 17 Numerical simulations of nanodroplet jumping. (a) First numerical simulations of coalescence-induced nanodroplet jumping, using same-sized Ar droplets on a nonwetting gold substrate. Reproduced with permission.¹³⁵ Copyright 2015, AIP Publishing. (b) Plot of v^* versus Oh for nanodroplets with varying gas/liquid viscosity ratios, for both molecular dynamics (MD) and volume-of-fluid (VoF) simulations. The results from (a) are overlaid as 'X' symbols, whose lack of dependence on Oh is in contrast to the broader numerical results shown here. The critical cutoff for jumping depended upon both Oh and μ_g/μ_l (inset). Reproduced with permission.¹³⁶ Copyright 2020, the Royal Society of Chemistry. (c) A more recent work used MD simulations to consider the inflation and wetting transitions before coalescence and jumping, with the final outcome depending on the relative positions of droplets with respect to the surface structure. Reproduced with permission.¹³⁷ Copyright 2023, Elsevier.

jumping efficiency was constant at $v^* = v_j/v_{ci} \approx 0.126$ and $\eta \approx 2\%$ over the entire data range, which is in good agreement with the lower limit of $v_{min}^* \approx 0.111$ for large Oh numbers predicted by Enright et al.²² using finite-element simulations. Given that v^* is constant over this size range, the actual jumping velocity continually increased with decreasing droplet size, to a record speed of $v_j \approx 3$ m/s at $D_i = 23$ nm.¹³⁵

Follow-on studies refined our understanding of nanodroplet jumping. Another MD study varied the liquid–solid bonding parameter to effectively vary the apparent contact angle and surface adhesion.¹³⁸ It was found that coalesced nanodroplets fail to jump when $\theta^* \approx 145^\circ$ but succeed when $\theta^* \approx 175^\circ$. A third MD study explicitly modeled the underlying nano-pillars, varying the solid fraction and surface wettability.¹³⁹ Intuitively, jumping was possible at lower solid fractions and wettabilities, whereas nonjumping occurred at larger solid fractions and wettabilities. In 2020, Perumanath et al.¹³⁶ cross-correlated MD simulations of nanodroplet jumping with volume-of-fluid continuum simulations (Figure 17b). They found not only that v^* varied with Oh after all, but additionally with the ambient gas pressure (i.e., drag during coalescence) and interfacial thermal fluctuations (causing trial-to-trial variations in the droplets' centers of mass). For a given surface, three nondimensional numbers can therefore affect the value of v^* : the Ohnesorge number as previously discussed, the Knudson number $Kn = \lambda/R$ (where λ is the mean-free-path of gas molecules), and the thermal fluctuation number $Th = \sqrt{k_B T/\gamma R^2}$ (where k_B is the Boltzmann constant and T is the temperature).

Very recently, Gao et al.¹³⁷ performed extensive large-scale MD simulations that included an array of superhydrophobic nanopillars and characterized the initial nucleation and growth processes that precede coalescence. This work demonstrated numerically that embryos preferentially nucleate within the texture, in agreement with classical nucleation theory. They demonstrated that confined growth could alternately result in single-droplet self-jumping,

Wenzel-to-Cassie transitions before or during coalescence-induced jumping, or coalescence into impaled Wenzel droplets, depending on the surface architecture and nucleation density (Figure 17c). The extension of self-jumping to nanodroplets seems practically important, as jumping requires extensive subcoolings of $\Delta T \gtrsim 10^\circ\text{C}$ to ensure that droplets coalesce at nano-scale sizes.

Variation #12: Alternative working fluids

The first demonstration of capillary-inertial jumping with low-surface-tension fluids was in Boreyko's 2012 PhD thesis, which showed that ethanol droplets colliding on a Leidenfrost stage do not jump as high as water droplets (Figure 18a,b).¹⁴⁰ In 2017, Vahabi et al.⁴⁰ demonstrated that both high-viscosity (e.g., water–glycerol mixtures) and low-surface-tension (e.g., water–SDS mixtures) liquids are capable of jumping from a superomniphobic mesh with re-entrant textures. By widely varying the volume fraction of glycerol, it was discovered that the jumping efficiency remains fixed at $v^* \approx 0.18$ regardless of the liquid's viscosity, provided that $Oh \ll 1$ (Figure 18c, green region). When the visco-capillary time scale first approaches the same order of magnitude as the capillary-inertial time scale, which occurred experimentally around $Oh \approx 0.1$, the value of v^* swiftly drops until jumping is no longer possible (yellow region). For example, $v^* \approx 0.1$ for $Oh \approx 0.3$ and $v^* \approx 0.01$ (i.e., final cutoff for jumping) at $Oh \approx 0.7$. From a purely analogous standpoint, the precipitous drop in v^* above a critical Oh number is reminiscent of the sudden drop in v^* beneath a critical surface-dependent droplet size for dew (previously discussed above),² but the mechanisms are completely different (viscous effects vs. an immature droplet inflation).

When using a macrotexture to enhance the jumping hydrodynamics, jumping was even observed for a 90% glycerol droplet ($\mu \approx 220$ mPa·s) with an Ohnesorge number larger than unity

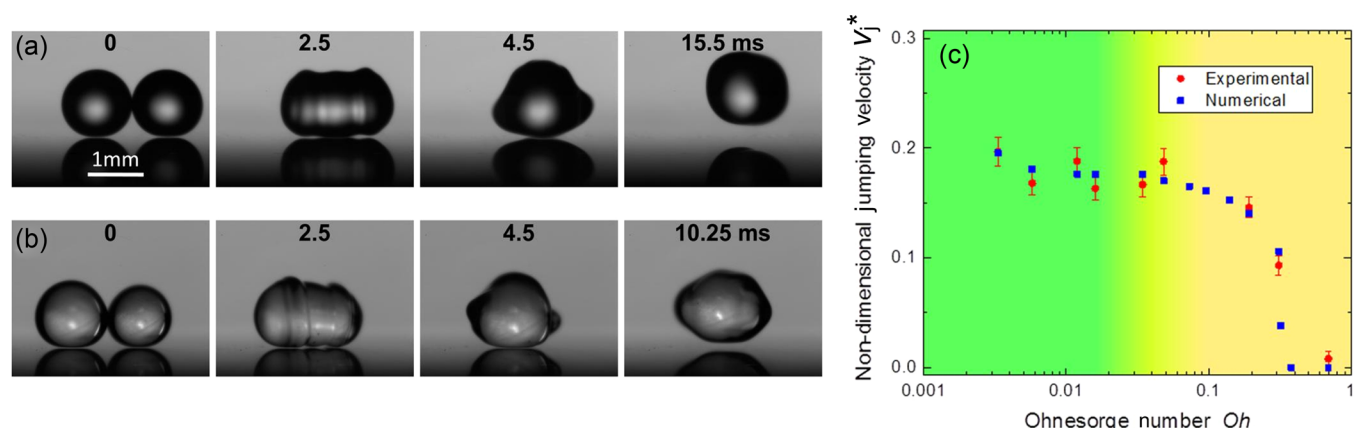


FIGURE 18 Effects of varying the surface tension and viscosity of the droplets. (a, b) Using a Leidenfrost droplet collider, (a) water droplets were observed to jump considerably higher than (b) ethanol droplets due to their higher surface tension.¹⁴⁰ (c) Varying the droplet viscosity with glycerol, the value of Oh could be widely varied to observe the decay of the jumping efficiency from $v^* \approx 0.2$ (pure water) to $v^* \rightarrow 0$ (85% glycerol, $\mu \approx 109$ mPa·s). Reproduced with permission.⁴⁰ Copyright 2017, the American Chemical Society.

($Oh \approx 1.14$).¹¹² Finally, a recent report by Hou et al.¹⁴¹ observed the coalescence-induced jumping of heterogeneous droplets, where one droplet was water, while the other was a water–glycerol mixture. During coalescence, the gradients in surface tension and viscosity result in an asymmetric distribution of internal pressure and a weak decrease in the energy conversion efficiency.

Variation #13: Jumping from smooth fibers

Thus far, all mentions of capillary-inertial droplet jumping have required a superhydrophobic surface or a highly superheated Leidenfrost stage. It was therefore rather shocking when Zhang et al.¹³² reported coalescence-induced jumping from smooth hydrophobic fibers in 2015 (Figure 19a,b). Compared to a planar substrate, a droplet on a fiber shows a larger volume-to-contact-area ratio. In other words, most of the droplet's area is interacting with the air, rather than the fiber. This serves to increase the apparent contact angle of a droplet on a smooth hydrophobic fiber, relative to its intrinsic angle on a smooth hydrophobic plane. It also serves to reduce the droplet–substrate contact area for a given droplet volume. For these reasons, microdroplets can spontaneously jump from Teflon-coated fibers upon coalescence without requiring a superhydrophobic Cassie state.

Interestingly, whereas the jumping efficiency is independent of droplet size for sufficiently inflated droplets on superhydrophobic surfaces,² here, the increased importance of the work of adhesion renders the jumping efficiency dependent on the droplet-to-fiber radius ratio (Figure 19c).¹³² This can be analytically expressed by the following semi-empirical equation:

$$v^* = \frac{V_i}{V_{ci}} = v_\infty^* \left(1 - \frac{\Gamma_c}{\Gamma} \right)^{1/2}, \quad (13)$$

where $\Gamma = R/R_{\text{fiber}}$ is the droplet-to-fiber radius ratio, v_∞^* is the value of v^* for an infinitely large value of Γ (where the work of adhesion is negligible), and Γ_c is the critical radius ratio above which merging

droplets can detach from the fiber. For water droplets on Teflon-coated fibers, $v_\infty^* \approx 0.7$ and $\Gamma_c \approx 3$, yielding an excellent fit to the experimental data for v^* versus Γ . For a large radius ratio of $\Gamma \approx 20$, high jumping efficiencies of $v^* \approx 0.7$ and $\eta \approx 50\%$ were measured, far larger than what is possible with superhydrophobic surfaces (in the absence of enhancing macrostructures). A follow-up report developed a more complex model that more explicitly balanced the kinetic energy, excess surface energy, work of adhesion, and viscous dissipation.¹⁴²

The key takeaway here is that while a droplet on a smooth hydrophobic surface intrinsically has a much larger work of adhesion (per area) than a Cassie droplet on a superhydrophobic surface, the use of slender fibers provides the benefit of greatly increased volume-to-contact-area ratios that out-competes the disadvantage of increased adhesion. Ironically, this improved jumping ability on smooth hydrophobic fibers is not necessarily attractive: for fog harvesting, it has been demonstrated to prevent droplet drainage and collection.¹⁴³

Variation #14: Delayed frost growth

For a moderately supercooled surface that is initially dry, it is rare for water vapor to directly desublimite into frost. This is because the energy barrier for nucleating a supercooled liquid embryo is slightly less than that of an ice embryo (except at ultra-low temperatures or for superhydrophilic surface chemistries).¹⁴⁴ For a hydrophobic or superhydrophobic surface, the result is a population of supercooled dew droplets that will eventually freeze into “mother” frost before diffusively growing dendritic “daughter” frost. Until the 2010s, the frost literature presumed that mother frost was formed by each supercooled dew droplet freezing in isolation at its liquid–solid interface, such that the droplets did not influence each other.^{145–149} A decade ago, this false notion was shattered by three independent reports of inter-droplet frost growth: a 2010 PhD thesis by Dooley,¹⁵⁰ a 2013 article by Boreyko and Collier,¹⁵¹ and a 2013 letter by Guadarrama-Cetina et al.¹⁵² These reports revealed that only a single

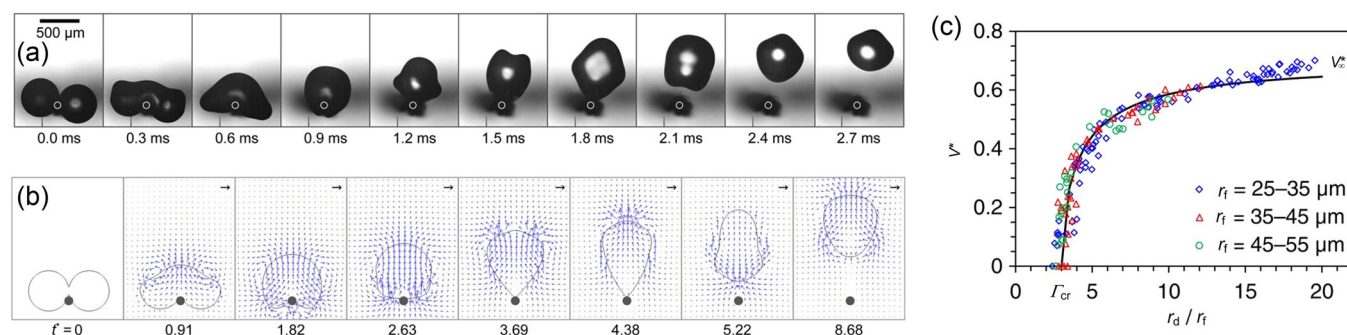


FIGURE 19 Jumping droplets on smooth fibers. (a, b) Experimental and numerical demonstrations of coalescence-induced jumping from a smooth, hydrophobic fiber. The reduction in the liquid–solid contact area offsets the increased wettability relative to a superhydrophobic surface. (c) The nondimensional jumping velocity varies widely depending on the ratio of the droplet radius (r_d) compared to the fiber radius (r_f). The black line represents a semiempirical energy balance model. Reproduced with permission.¹³² Copyright 2015, the American Physical Society.

supercooled droplet has to freeze from its contact with the chilled substrate, at which point, a network of interdroplet ice bridges percolates across the entire dew population. The mechanism involves the hygroscopic character of ice, which imparts a depressed saturated vapor pressure relative to supercooled liquid water at the same temperature.¹⁵³ This results in the diffusive evaporation of adjacent liquid droplets (even in a highly supersaturated ambient!); the harvested vapor deposits as ice bridges extending from the frozen droplet toward the liquid droplet(s).^{154,155} A schematic of diffusive inter-droplet ice bridging is shown in Figure 20a.¹⁵¹

Jumping-droplet condensation is highly relevant to this new framework of incipient frost growth, as its supercooled droplet size distribution is very dilute relative to that of dropwise condensation at the onset of freezing (Figure 20b).¹⁵¹ This serves to increase the time required for ice bridges to successfully grow to their liquid neighbors to freeze them in turn.¹⁵¹ Indeed, many micro-droplets completely evaporate before a growing ice bridge has the chance to connect, curtailing the chain reaction and locally drying the surface instead. When the liquid droplet is smaller than the frozen droplet, by conservation of mass, it should completely evaporate (rather than freeze) for a nondimensional bridging parameter $S^* \sim L_{\max}/D_l > 1$, where D_l is the initial diameter of the liquid droplet and $L_{\max}$ is the maximum possible bridge length that extends from the edge of the frozen droplet to the center of the evaporating liquid droplet.^{151,154} When the liquid droplet is larger than the frozen droplet and associated ice bridge, the drying criterion switches to $S^* > (D_l/D_f)^2$.¹⁵⁴ Evaporated droplets result in dry zones showing a characteristic length scale of $\delta \sim 100\mu\text{m}$ and can be maintained indefinitely, as fresh embryos

nucleating within the dry zone show a net negative vapor flux and spontaneously disappear.¹⁵⁶ Figure 20c shows that the S^* values are considerably higher for a jumping-droplet surface compared to a dropwise condenser, resulting in a threefold reduction in the radial propagation speed of condensation frosting.¹⁵¹ Inter-droplet ice bridging can be further impaired by micro-pillars when using a hierarchical superhydrophobic surface¹⁵⁷ or by using a biphilic surface.¹⁵⁸ Inspired by the Namib desert beetle or the veins of leaves, antifrosting surfaces can also be promoted with micro-patterned surfaces, where overlapping dry zones are sustained between sacrificial arrays of dilute ice features.^{159,160} For more details, the reader is referred to a recent book chapter on condensation frosting.¹⁶¹

Variation #15: Jumping bubbles

What if the liquid and gas phases of capillary-inertial jumping were inverted? Surprisingly, the concept of coalescence-induced bubble departure was not reported until a full decade after the 2009 report on jumping droplets. In 2018, Soto et al.¹⁶² used a supersaturated solution of CO_2 -in-water to induce the diffusive growth of CO_2 surface bubbles on a silicon substrate (Figure 21a). By patterning the hydrophilic silicon chip with hydrophobic micro-pits, the gas bubbles reliably nucleated within the pits to enable same-sized coalescence events. As with the liquid bridge for jumping water droplets, the expansion of the gas bridge was confirmed to be capillary-inertial in nature. The subsequent jumping event only launched a merged bubble about $h \approx 10\mu\text{m}$ from the surface,¹⁶² compared to $h \sim 1\text{mm}$

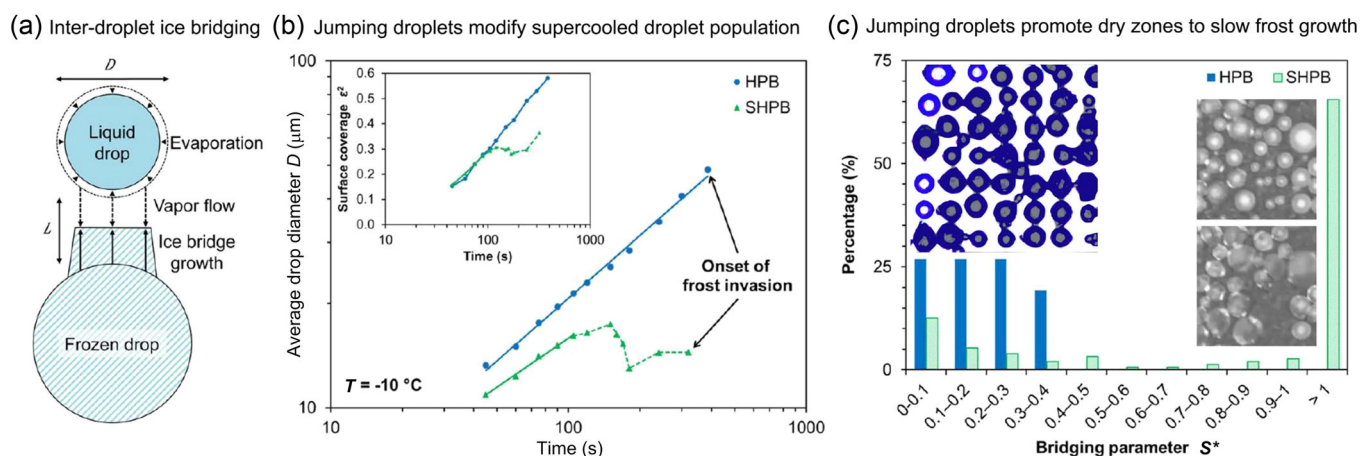


FIGURE 20 How jumping-droplet condensation retards frost growth. Reproduced with permission.¹⁵¹ Copyright 2013, the American Chemical Society. (a) When one supercooled dew droplet freezes (bottom), it becomes hygroscopic to diffusively harvest water vapor from neighboring liquid dew (top). This diffusive harvest manifests as a depositing ice bridge that eventually connects to and freezes the liquid droplet, setting off a chain reaction of inter-droplet frost growth. (b) The average droplet diameter increases steadily for dropwise condensation (blue circles) but levels off for jumping-droplet condensation (green triangles). For a chilled surface with supercooled condensation, this results in a more dilute droplet population at the onset of freezing. (c) Distribution of the nondimensional bridging parameter (S^*) during inter-droplet frosting for hydrophobic (blue) and superhydrophobic (green) surfaces. For the dropwise condensation, $S^* < 1$, such that inter-droplet bridging is always successful and all droplets freeze. This is depicted in the left inset, where the frozen droplets are false-colored and it can be seen that ice bridges easily connect all droplets together. For jumping-droplet condensation, $S^* > 1$ most of the time, such that many droplets completely evaporate before ice bridges can connect. This is shown in the right insets, where several droplets in the bottom-right corner of the top image are completely evaporated in the bottom image after 2 s of subsequent frosting has occurred.

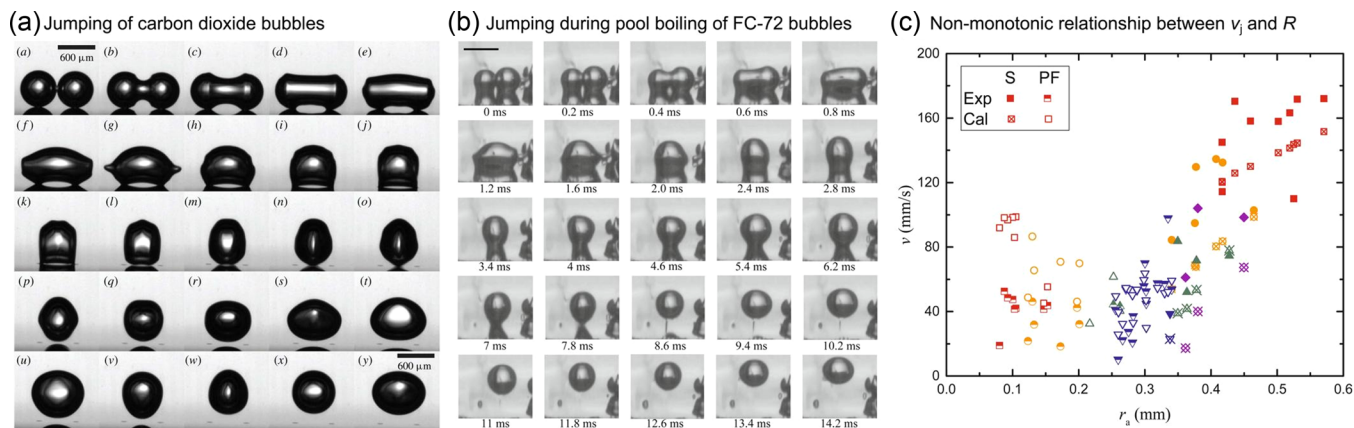


FIGURE 21 Buoyancy-assisted jumping bubbles, as an emerging analog to jumping droplets. (a) First side-view capture of coalescence-induced jumping bubbles, using diffusively grown CO₂. Micro-cavities were lithographically etched into silicon to control the contact area and inter-bubble spacing before coalescence. The jump itself was relatively modest, but after detachment, buoyancy drives continued liftoff. Reproduced with permission.¹⁶² Copyright 2018, Cambridge University Press. (b) First demonstration of coalescence-induced jumping during pool boiling, used a refrigerant (FC-72) as the working fluid. Micropillars were used to help facilitate controlled coalescence and detachment events. (c) Graph of v_j versus R for the results from (b). Note the nonmonotonic behavior, due to a new regime on the right-hand side where v_j increases with R , opposite to the capillary-inertial trend. Figures (b, c) reproduced with permission.¹⁶³ Copyright 2020, Elsevier.

for jumping droplets.^{2,37,46} This poor quality of jumping is due to the lower surface tension, increased density and viscosity of the outer phase, and increased surface adhesion relative to launching water from a superhydrophobic surface. Nonetheless, buoyancy facilitates the continued departure of jumped bubbles (a convenience not afforded to jumping droplets!).¹⁶² A follow-on work by the same group analyzed the viscous dissipation inherent to the coalescence-induced departure of hydrogen peroxide bubbles, finding a critical bubble size above which departure is possible.¹⁶⁴

The first report of coalescence-induced bubble departure for a pool boiling system was in 2020 by Zhou et al.,¹⁶³ using FC-72 as the working fluid. The silicon substrate contained an array of square micro-pillars, which helped to control the nucleation sites and promote similar-size coalescence events (Figure 21b). Interestingly, the jumping departure speed varied nonmonotonically with bubble radius, with v_j initially decreasing with increasing R but then increasing with R for bubble radii exceeding 250 μm (Figure 21c). This is completely opposite to the nonmonotonic behavior of jumping water droplets, where v_j initially increases with R due to the increasing maturity of droplet inflation, followed by a decrease with R according to the capillary-inertial velocity scaling.

Very recently, we found the same nonmonotonic trend as Zhou et al.,¹⁶³ but in the context of the pool boiling of water on an aluminum surface with rationally structured micro-cavities.¹⁶⁵ We suspect that inflation is not responsible for either regime for jumping gas bubbles, as the observed bubbles were all fairly large ($R \gtrsim 100$ μm) and well-inflated before coalescence. Instead, we hypothesize that the first regime for bubbles is the well-known capillary-inertial jumping regime: $v_j \sim \sqrt{\gamma/(R\rho_l)}$ (where ρ_l is the density of the liquid ambient), as supported by high-speed side-view imaging which confirmed the vapor bridge impacting into the substrate. For the second regime, which occurred for water vapor bubbles above $R \gtrsim 300$ μm, we observed that

the buoyant force is now strong enough to prevent the vapor bridge from appreciably impacting against the substrate. In other words, coalescence still induces neck detachment, but not jumping. We term this bubble-exclusive departure regime the buoyant-inertial regime, which balances the buoyant energy ($E_b \sim \rho_l R^4 g$) against the kinetic energy:

$$v_{bi} \sim \sqrt{g} R^{1/2}, \quad (14)$$

which is in agreement with our experimentally measured power law for larger micro-bubbles $v_j \propto R^{1/2}$. The efficiency of buoyant-inertial departure is very low, with $v^* = v_d/v_{bi} \approx 0.03$ (where v_d is the actual departure speed), most likely due to the energy lost to depinning the vapor necks during coalescence. A recent numerical work by Iwata et al. used numerical simulations to estimate the critical size for the coalescence-induced departure of vapor bubbles from a smooth wall, as a function of contact angle and nucleation density.¹⁶⁶

CONCLUSION

In summary, when two or more micro-droplets (or bubbles) coalesce together on a substrate that promotes large apparent contact angles and low surface adhesion, the release of surface energy is partially converted into kinetic energy that manifests as out-of-plane jumping. The mechanism for the out-of-plane directionality is the symmetry breaking that occurs when the expanding liquid bridge impacts the substrate on one side but expands freely on the other side. The primary context for this phenomenon is jumping-droplet condensation on a superhydrophobic surface, although several studies deposit micro-droplets to facilitate cleaner imaging and superior control of droplet size and placement. Upon coalescence, the majority of the

released surface energy is converted into oscillatory kinetic energy, with only about 4% being successfully converted into translational kinetic energy. As a result, the initial jumping speed upon coalescence is roughly one fifth that of the idealized capillary-inertial velocity. Remarkably, surface tension is so powerful at the micro-scale that dramatic jumping speeds of $v_j \sim 0.1$ m/s and against-gravity heights of $h \gtrsim 1$ mm are typical, despite the ultra-low efficiency. During coalescence and the initial jumping, some energy is also lost to contact angle hysteresis, work of adhesion, and nonoscillatory viscous dissipation, although for micro-droplets on a robust superhydrophobic surface, these losses are quite small relative to the oscillatory kinetic energy. After launching, the translational kinetic energy is primarily dissipated by air and/or vapor drag, whereas the period-averaged oscillatory kinetic energy is weakly dampened by internal viscous dissipation.

Benchmarks for characterizing the hydrodynamics of jumping-droplet condensation

We suggest that future experimental or numerical studies on capillary-inertial jumping should report three key benchmarks: the critical droplet size required for coalescence-induced capillary-inertial jumping to be possible (R_c), the nondimensional initial jumping speed ($v^* = v_j/v_{ci}$), and the energy conversion efficiency ($\eta = E_{k,tr}/\Delta E_s$). The first metric, R_c , is a measure of the robustness of the underlying superhydrophobic surface and is *not* related to a critical Ohnesorge number as wrongly hypothesized by many reports. Surfaces with smaller values (ex: $R_c \sim 1$ μ m) promote early droplet inflation to a mature dew droplet whose outer contact line shows a large apparent contact angle and low surface adhesion (i.e., air-trapping Cassie state). This enables a high likelihood of coalescence-induced jumping upon coalescence, especially if the droplets are similar-sized. Tall and slender nanopillars, or especially a close-packed array of nanocones, have been shown to promote ultra-low values of R_c for reliable droplet jumping. Conversely, surfaces with large values ($R_c \gtrsim 10$ μ m) result in more coalescence events where jumping does not occur, as more droplets that are coalescing remain primarily impaled within the surface structure. Shorter or more dilute nanostructures result in larger values of R_c . For experimental work utilizing deposited droplets, or numerical simulations that presuppose mature droplet inflation, R_c is still relevant but now corresponds to a much smaller length scale ($R_c \sim 10$ nm for water), representing the viscous cutoff limit for jumping at a critical Ohnesorge number. The concept of a critical jumping radius is still useful in this viscous context, as the critical Ohnesorge number (and by extension R_c) has already been shown to vary anywhere from $0.3 < Oh_c < 1.2$ depending on the surface structure, additional presence of a macrostructure, and/or choice of working fluid. In the future, researchers should further rationalize the true lower limit of R_c for a variety of different surface types and working fluids; for example, perhaps a rig could be designed that is capable of experimentally capturing the launch of (fluorescently dyed?) jumping nanodroplets for the first time.

The second and third metrics, v^* and η , are both tied to the hydrodynamic efficiency of jumping. We suggest that v^* be obtained by normalizing the measured initial jumping speed by a universal capillary-inertial length scale: $v_{ci} \sim \sqrt{\gamma/(\rho R)}$. This allows for graphs of v^* versus R where every data point is nondimensionalized in the same manner, regardless of trial-to-trial variations in size mismatch or contact angle of the droplets. This also makes it easy to directly compare values of v^* across the literature, although obviously direct comparisons should only be made for similar contact angles and size mismatch conditions. Subsequently, when converting into a graph of η versus R , caution should now be exercised to account for how the varying contact angles and radii of the droplets affect the precise amount of released surface energy, ΔE . In other words, the numerical prefactor in the expression $\Delta E \propto \gamma R^2$ will be slightly different for each data point to ensure an accurate estimation of the energetic conversion efficiency from released surface energy into translational kinetic energy. Given the frequent reports of record-breaking values of v^* and η nearly every year, it is clear that the true performance ceiling for capillary-inertial jumping has not yet been reached and more innovations are sure to follow. The use of macrotextures to increase jumping efficiency is the temporal inverse of using superhydrophobic macrotextures to reduce the contact time of a rebounding droplet, by splitting it into satellite droplets.^{167,168}

Benchmarks for characterizing the heat transfer of jumping-droplet condensation

In terms of phase-change heat transfer, there are at least three different methods for measuring h for a given jumping-droplet condenser. The most direct way is to measure the surface heat flux for a given subcooling, either using an array of thermocouples or by measuring the flow rate and temperature increase of the coolant passing through the heat sink. A second tactic is to measure the mass flow rate of dew shedding from the condenser, which can be converted into heat transfer using: $h \approx \dot{m} L_v / (A_{\text{cond}} \Delta T)$, where L_v is the latent heat of vaporization, A_{cond} is the condenser area, and ΔT is the surface subcooling with respect to the steam. A third, and increasingly common tactic, is to quantify the steady-state droplet size distribution of a jumping-droplet condenser and extract the value of h using an analytical or numerical heat flux model.

To our knowledge, no study has directly compared all three techniques for quantifying phase-change heat transfer for the same condenser and system conditions. This cross-correlation could be quite impactful, especially considering the increasing number of differing models for how to quantify the droplet size distribution and associated heat flux for a population of jumping dew droplets. For example, perhaps the direct measurement approach could be used as the baseline case to quantify the percent error of the mass measurement approach and the various approaches to modeling the thermal resistance of a rationalized droplet population. We also expect that advances in machine learning will enable automated and highly accurate measurements of the droplet shedding rate and

droplet size distribution to assist in calculating heat transfer coefficients.¹⁶⁹

An under-appreciated fact is that the droplet size distribution, shedding rate, and heat transfer are all tied to the surface orientation of the jumping-droplet condenser. This is because jumped droplets tend to return and eventually get stuck at more horizontal orientations, whereas inclined orientations can promote coalescence-induced detachment from the Wenzel necks to foster droplet sliding and sweeping, to complement jumping removal.⁹⁰ Therefore, in the future, researchers should measure heat transfer at multiple surface orientations and only cross-compare to other surfaces in the literature for an equivalent orientation.

For the enhanced phase-change heat transfer and/or antifogging of jumping-droplet condensation to be worthy of sustained academic interest, serious thought should be given to pathways for practical implementation. Eventually, a breakthrough in surface chemistry will be required to render superhydrophobic nanostructures and coatings durable under long-term exposure to harsh steam. An encouraging example is the study of Donati et al.,¹⁷⁰ who used a carbon nanofiber composite coating to preserve jumping-droplet condensation for roughly 10 h under very harsh steam flow conditions. In the future, researchers should aspire to develop coatings that last for ~100 h or even ~1000 h to demonstrate industrial relevance. Alternatively, applications should be identified that only require short-term performance or allow for periodic surface rejuvenation.

Avenues for future research on capillary-inertial jumping

As the jumping-droplet condensation of water on a superhydrophobic nanostructure is now a fairly mature field of study, researchers are especially encouraged to consider alternative contexts to capillary-inertial jumping. For example, the use of low-surface-tension fluids for jumping has been confined to depositing droplets atop a re-entrant surface structure; is it possible to design an omniphobic surface that could successfully inflate large-angle dew droplets for low-surface-tension liquids growing within the surface structure? This would presumably require a hierarchy of re-entrant textures embedded within each unit cell of roughness, which would be an interesting challenge for those specializing in nano-fabrication. Or is it possible to engineer a superomniphobic nanostructure where the vapor cannot penetrate into the roughness in the first place?

Another topic worthy of further study is droplet jumping from smooth micro-fibers. This approach does not require fragile superhydrophobic coatings and is potentially relevant for applications including condensation, fog harvesting, screening of aerosols containing COVID-19 or other viruses, and more. In the context of fog harvesting or screening aerosols, the jumping of captured droplets from the fibers is usually undesirable, so research efforts could focus on minimizing the effect rather than promoting it.

The use of micro- or macro-structures to promote directional jumping and sweeping chain reactions is gaining research momentum

since its discovery in 2015, but could use further development in terms of quantifying and optimizing the speed and propagation length scale of the cascade effects. Single-droplet jumping, via an asymmetric Laplace pressure, is also a highly promising avenue of future study, as it decouples droplet launching from a dependence on coalescence and nucleation density. Several recent reports have shown dramatic gains in the robustness of single-droplet jumping, so clearly, there are many innovations that still remain to be made in this direction.

There is a direct connection between maximizing the jumping- or sweeping-droplet removal from a surface and its subsequent delay in inter-droplet condensation frosting; perhaps with enough optimization, a chilled surface could universally lead to dry zones about any frozen droplet to minimize its frost coverage. The discovery of jumping-droplet condensation on wheat leaves and a variety of animals, and its important implications for governing the rate of disease spread or self-cleaning, should encourage more collaborations with agriculturists, plant pathologists, and biologists. Surely there are many more organisms where jumping-droplet condensation is playing an important ecological role, completely unbeknownst to us.

Finally, the recent discovery of the coalescence-induced departure of surface micro-bubbles has profound implications for enhancing boiling heat transfer. While most studies on optimizing the critical heat flux of a boiler have focused on increasing the wickability of the superheated surface,¹⁷¹ it is very likely that the coalescence-induced tuning of the critical bubble departure size is of equal importance, and a synergistic combination of these tactics could result in unprecedented heat fluxes. Indeed, we expect that jumping bubbles have more promising industrial prospects than jumping-droplet condensation, as the former only requires a large and hydrophilic micro-texture to controllably promote, whereas the latter requires a nanotexture with a conformal hydrophobic coating, which is expensive and fragile. Bubbles can controllably nucleate and inflate within a large micro-feature because their critical embryo size is several orders of magnitude larger than that of a liquid embryo,⁶⁰ while large contact angles are possible with bubbles for an uncoated hydrophilic surface (i.e., a low contact angle for the water phase is desirable). Jumping bubbles are also not dependent on achieving a large energy conversion efficiency, as buoyancy can take over the departure kinematics after the initial coalescence-induced detachment from the surface. Fundamentally, jumping gas bubbles are very rich, as this construct allows for a wide variety of working fluids, modes of bubbles growth, and the addition of buoyancy into the hydrodynamics. Perhaps it is therefore time for researchers in this field to make their own jump, from the framework of liquid-phase droplets to gas-phase bubbles, in order for capillary-inertial jumping to reach new heights.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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